

VABench: Measuring Embodied Spatial Intelligence through Visual Demonstrations, Active Perception, and Metric Control

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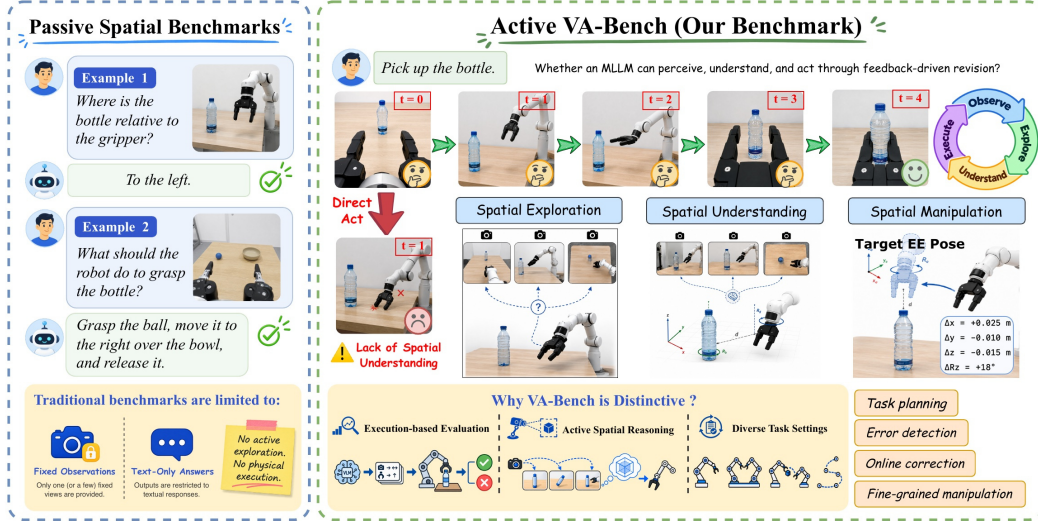


Figure 1: Overview of the evaluation target. Fixed-view spatial evaluations can stop at a verbal description, whereas VA-Bench requires an MLLM to actively acquire evidence, infer robot–object geometry, issue metric manipulation commands, and revise them from execution outcomes.

ABSTRACT

Spatial intelligence requires more than describing object locations. Under incomplete observation, models must identify and acquire missing evidence, interpret it in a common spatial frame, and act on it. We introduce VA-BENCH to evaluate the complete observe–reason–act–revise loop. General-purpose MLLMs learn procedural context from RGB-only demonstrations, actively select camera view-points, issue metric Cartesian commands, and revise them from execution feedback. Models receive no privileged object poses, oracle trajectories, or learned action heads. A fixed model-agnostic controller executes only model-specified targets. VA-Bench contains 14 base task families (11 single-arm and three dual-arm), seven held-out geometry/layout variants, and a long-horizon five-object composition track. We evaluate 12 primary model conditions in three independent runs over the same 20 physically verified seeds per base task, reporting terminal success, nine trajectory-level behavioral diagnostics, and subtask progress. **First, the best-performing model scores 100.0% on target localization and 78.9% on spatial relations in the annotated run. Its three-run macro-average task success is only $53.93 \pm 3.17\%$. Second, active camera control significantly improves task success over passive multi-view observation. In one matched comparison, success rises from 27.86% to 57.50%. Third, held-out geometric transfer can reduce task success by over 30 percentage points. No model completes a strict long-horizon episode, despite substantial partial progress.** VA-Bench provides a multi-task stress test for general-purpose MLLMs as embodied agents. It evaluates whether they can apply task constraints inferred from visual demonstrations, actively acquire task-relevant visual evidence, and use spatial understanding to complete manipulation tasks in physics-based simulation.

1 INTRODUCTION

Spatial intelligence is not limited to describing where objects appear in an image. Under partial observability, an agent must determine what information is missing, acquire it, and interpret it in a common spatial frame before acting (Liu et al., 2023b; Kamath et al., 2023; Du et al., 2024; Song et al., 2025). This requirement is especially important for manipulation: viewpoint, distance, and occlusion can make the metric relationship between the robot and an object ambiguous in a single view, and pixel displacement alone does not determine motion in world coordinates (Fu et al., 2024; Yang et al., 2025a; Ma et al., 2025; Azuma et al., 2022; Ma et al., 2023). Reliable interaction therefore depends on active evidence acquisition and contact-level spatial estimation (Chen et al., 2024; Cheng et al., 2024; Liao et al., 2024; Nasiriany et al., 2024b; Yuan et al., 2025; Huang et al., 2025).

Recent benchmarks extend spatial evaluation from fixed- and multi-view reasoning to embodied planning (Fu et al., 2024; Yang et al., 2025a; Song et al., 2025; Zheng et al., 2022; Zhao et al., 2025; Zhang et al., 2025; 2026; Hong et al., 2026; Gao et al., 2026; Yang et al., 2025b; Wu et al., 2026; Maurya et al., 2026). They test important components of spatial intelligence. Evidence acquisition, inference, and execution, however, remain coupled. Each observation or action changes the evidence for the next decision (Huang et al., 2023b; Nasiriany et al., 2024b; Huang et al., 2025; Zhang et al., 2026; Hong et al., 2026). This motivates our central question: *when a model must close this loop on its own, can spatial understanding support reliable, executable action under the constraints of physical interaction?*

Figure 1 illustrates the evaluation gap that motivates VA-Bench. A model may describe a spatial relation correctly without estimating geometry precisely enough for contact (Chen et al., 2024; Liao et al., 2024; Nasiriany et al., 2024b; Yuan et al., 2025), while preselected views do not reveal whether it can identify and resolve missing evidence (Fu et al., 2024; Yang et al., 2025a; Ma et al., 2025; Zhang et al., 2026; Hong et al., 2026). Likewise, predefined skills or policies may execute a task without testing whether the model can convert spatial judgments into metric robot motion (Ichter et al., 2023; Huang et al., 2023b; Liang et al., 2023; Driess et al., 2023; Rana et al., 2023; Huang et al., 2023a; Nasiriany et al., 2024b; Fang et al., 2024; Huang et al., 2025). VA-Bench evaluates these capabilities jointly within a single closed-loop interaction.

We introduce VA-BENCH, a physics-simulated benchmark for this complete observe–reason–act–revise loop. Given RGB observations and task instructions, general-purpose MLLMs select viewpoints and produce metric Cartesian commands. They receive no privileged object poses, oracle trajectories, or task waypoints. A fixed model-agnostic controller only executes the targets specified by the MLLM. A successful RGB demonstration supplies procedural context but no geometry for the current test instance. The evaluation therefore includes both procedure interpretation and instance-specific re-grounding.

VA-Bench contains 14 task families, including 11 single-arm and three dual-arm families. They span grasping, placement, tool use, handover, and coordinated manipulation. Each primary condition is evaluated in three independent runs over the same 20 deterministic, physically verified seeds per task, yielding 840 run–seed episodes scored by terminal task success or failure. We further test controlled geometry and layout transfer and a five-object long-horizon task. Nine trajectory-level behavioral diagnostics and subtask-progress measures localize observable degradation along the interaction loop without attributing every failure to spatial reasoning alone.

Our evaluation of 12 primary model conditions yields three main findings. **First, strong target localization does not ensure reliable task execution.** The best-performing model attains 100.0% on target localization and 78.9% on spatial relations in the annotated run. Its three-run 14-task macro-average success is only $53.93 \pm 3.17\%$. **Second, active camera control improves task success over passive multi-view observation.** In a matched comparison, success rises from 27.86% to 57.50% with active camera control. Across all 12 conditions, the best active-exploration diagnostic score is 83.2%. **Third, held-out geometric transfer and long-horizon composition remain challenging.** Across seven matched families, transfer degradation varies substantially: one condition drops 32.14 points, from 80.00% to 47.86%, while another drops 10.00 points, from 77.86% to 67.86%. The strongest conditions complete 61/100 and 53/100 object placements, respectively, but neither completes a single strict long-horizon episode.

Our contributions are threefold:

- We introduce VA-BENCH, which evaluates whether a general-purpose MLLM can interpret an RGB demonstration, actively acquire evidence, ground it into metric single- or dual-arm actions, and revise those actions from execution outcomes.
- We construct 14 task families with controlled held-out variants and a long-horizon compositional task, together with nine behavioral diagnostics and subtask-progress measures for localizing failures within the closed loop.
- We evaluate 12 primary model conditions over three runs and characterize three limitations of current closed-loop competence: the mismatch between spatial diagnostics and execution, degradation under geometric transfer, and failure to compose atomic progress into long-horizon completion.

2 RELATED WORK

2.1 SPATIAL REASONING UNDER PARTIAL OBSERVATION

Spatial reasoning has been studied extensively using static images, supplied multi-view observations, and 3D scene representations. CLEVR and GQA established benchmarks for compositional visual reasoning (Johnson et al., 2017; Hudson & Manning, 2019). SpatialVLM develops qualitative and quantitative spatial understanding (Chen et al., 2024), while CV-Bench, introduced with Cambrian-1, and subsequent evaluations probe spatial relations, depth, distance, and scene understanding in multimodal models (Tong et al., 2024; Fu et al., 2024; Yang et al., 2025a; Ma et al., 2025). ManipBench brings this diagnosis closer to manipulation by testing low-level reasoning from supplied observations, without an interactive execution loop (Zhao et al., 2025).

Theory of Space, ESI-Bench, and SpatialWorld make observation gathering interactive, evaluating spatial beliefs or task decisions as agents explore partially observed environments (Zhang et al., 2026; Hong et al., 2026; Gao et al., 2026). Their evaluations focus primarily on spatial beliefs, semantic decisions, or high-level task outcomes, rather than directly grounding acquired evidence into metric end-effector control with execution-level feedback. VA-Bench bridges this gap by requiring the same model to acquire task-relevant visual evidence, translate spatial understanding into metric arm motions, and revise subsequent decisions from execution outcomes.

2.2 EMBODIED MANIPULATION AND ROBOT BENCHMARKS

Meta-World, RLBench, CALVIN, LIBERO, and ManiSkill provide standardized environments for evaluating manipulation learning and generalization (Yu et al., 2020; James et al., 2020; Mees et al., 2022; Liu et al., 2023a; Mu et al., 2021; Gu et al., 2023); BEHAVIOR-1K and RoboCasa extend evaluation to diverse household activities (Li et al., 2023; Nasiriany et al., 2024a). A complementary policy-learning line, including RT-1, RT-2, OpenVLA, and Octo, learns robot action generation from demonstration data (Brohan et al., 2023; Zitkovich et al., 2023; Kim et al., 2025; Ghosh et al., 2024). These results establish strong visuomotor control capabilities, but their evaluation generally assumes learned policy execution or specialized action-generation modules rather than requiring a general-purpose MLLM to close the observation-to-action loop.

More directly related benchmarks already test general-purpose MLLMs in executed manipulation. EmbodiedBench’s manipulation setting supplies a fixed front view with object annotations and 3D coordinates (Yang et al., 2025b). ST-BiBench includes continuous bimanual control but relies on auxiliary ground-truth pose information (Wu et al., 2026). IMBench’s action stage evaluates both a vision-based agent and an object-state-augmented agent through parameterized motor primitives rather than direct metric action generation (Maurya et al., 2026). ESPIRE evaluates localization and manipulation execution through generated spatial outputs, with depth-based lifting from image coordinates to robot poses (Zhao et al., 2026). These benchmarks provide important progress toward executable manipulation evaluation. However, they typically rely on structured spatial information, object states, or predefined execution components. VA-Bench instead evaluates whether the same general-purpose MLLM can actively select observations and ground them into metric arm commands within a joint view–arm loop, without supplied object geometry or a learned robot action head.

2.3 ACTIVE PERCEPTION AND DEMONSTRATION-GROUNDED CONTROL

Active perception and next-best-view planning study how sensing actions reduce uncertainty or improve geometric coverage (Bajcsy, 1988; Pito, 1999). Their integration with manipulation also has direct precedents: Vision in Action learns task-relevant camera behavior and bimanual control from human demonstrations (Xiong et al., 2025b), while SaPaVe coordinates learned camera and arm policies and introduces ActiveManip-Bench for active manipulation (Liu et al., 2026). These works establish the value of joint sensing and control through learned visuomotor policies; VA-Bench examines this coordination through explicit decisions made by a general-purpose MLLM.

Demonstration-based learning likewise provides established routes from observed behavior to action. RoboMimic studies learning from recorded robot demonstrations, BC-Z uses language or video task conditioning, and MimicPlay combines human play videos with robot demonstrations (Mandlekar et al., 2022; Jang et al., 2022; Wang et al., 2023). MimicGen scales training data by adapting demonstration segments to new scenes (Mandlekar et al., 2023). These systems use action-bearing robot data to train control policies. In contrast, VA-Bench treats demonstrations as contextual guidance rather than training signals, requiring the evaluated model to infer reusable procedures from visual demonstrations and re-ground them in unseen scenes. In VA-Bench, the evaluated model first extracts a textual procedure from sampled RGB demonstration frames; execution must instantiate that procedure in the current scene without demonstration trajectories, object poses, or task-specific policy training.

Together, these lines motivate the setting summarized in Table 1: a *joint view–arm loop for general-purpose MLLMs without learned robot action heads*. VA-Bench studies whether general-purpose MLLMs can integrate demonstration-derived procedures, active observation, and metric manipulation within a unified closed-loop protocol.

Table 1: Comparison of representative evaluation settings. Entries describe the cited protocols, not every capability of their underlying simulators.

Setting	Active observation	Executed arm control	Joint view–arm loop	General MLLM, no learned head
ManipBench	No	No	No	Yes
Theory of Space	Yes	No	No	Yes
ESI-Bench	Yes	No	No	Yes
SpatialWorld	Yes	No	No	Yes
EmbodiedBench (manipulation)	No	Yes	No	Yes
ST-BiBench (fine control)	No	Yes	No	Yes
IMBench (action stage)	No	Yes	No	Yes
ESPIRE (execution)	No	Yes	No	Yes
ActiveManip-Bench / SaPaVe	Yes	Yes	Yes	No
VA-Bench	Yes	Yes	Yes	Yes

Active observation means explicit view-acquisition decisions, including navigation; incidental camera motion is excluded. *Executed arm control* includes parameterized motor primitives and geometrically lifted predictions, but excludes spatial QA and symbolic object interactions. A *joint view–arm loop* interleaves viewpoint and arm-control decisions within manipulation. The last column permits fixed controllers and does not imply identical observation inputs. ActiveManip-Bench is shown with SaPaVe’s learned-policy setting.

In contrast to prior benchmarks that evaluate spatial understanding, manipulation execution, or active perception separately, VA-Bench studies their interaction through a joint view–arm loop within a single closed-loop protocol, where an MLLM must acquire missing evidence, ground spatial concepts into metric actions, and adapt behavior based on physical feedback.

3 VA-BENCH: BENCHMARK DESIGN AND EVALUATION PROTOCOL

VA-BENCH, the *Vision-Action Benchmark*, evaluates general-purpose MLLMs as embodied agents in physics-based simulation. In the standard protocol, a model infers task context from visual demonstrations, acquires visual evidence, generates metric robot commands, and revises them from execution feedback. Each evaluation episode is scored by a physics-based task-success predicate.

Table 2: Coverage of the 14 VA-Bench base task families. Counts are the number of families in each analysis group.

Task group	No.	Primary spatial and manipulation demand
Direct grasp grounding	5	Locate a graspable region, approach it with the correct end-effector orientation, and close at a usable height/depth.
Placement and stacking	4	Track object–container or object–support relations and control release height, orientation, and final contact.
Functional contact and tool use	2	Select a contact point and approach direction for pressing, striking, or other goal-directed interaction.
Bimanual coordination	3	Assign arm roles, maintain relative geometry, and sequence synchronized or handover actions.

3.1 EVALUATION TRACKS AND TASK SUITE

Base track. The base suite contains 14 task families, including 11 single-arm and 3 dual-arm families, with 20 physically verified scene seeds per family. Table 2 groups them by their primary spatial and manipulation demands. Each family receives equal weight in the base score.

Controlled transfer track. Seven single-arm families have held-out variants that change object geometry, the container or box instance, or the layout while preserving the instruction. They test re-grounding of a learned procedure in a changed scene.

Long-horizon composition track. The five-object basket task combines four atomic manipulation skills into a longer sequence of placements that requires sustained progress tracking. Its demonstration structure and horizon differ from the base suite, so we report it separately.

All three tracks allow model-directed viewpoint selection. The passive-view control removes this choice and is reported as a separate ablation (Section 5.1).

3.2 INTERACTIVE EPISODE PROTOCOL

Each episode specifies a task family τ , a fixed scene seed, a task instruction, demonstration context, an action budget B_τ , and a success predicate ϕ_τ . The seed determines the scene but provides no privileged geometry to the model. Appendix C.1 gives the full instance specification.

Let x_t denote the physical simulator state. At decision step t , the observation consists of the current active-camera image I_t , filtered robot proprioception p_t , interaction history h_t , and the previous action/planner outcome f_{t-1} . The robot and camera action sets are $\mathcal{A}_{\text{robot}}$ and $\mathcal{A}_{\text{cam}}$:

$$o_t = (I_t, p_t, h_t, f_{t-1}), \quad \begin{cases} x_{t+1} = T_\tau(x_t, a_t), & a_t \in \mathcal{A}_{\text{robot}}, \\ x_{t+1} = x_t, & a_t \in \mathcal{A}_{\text{cam}}. \end{cases} \quad (1)$$

Robot actions change the physical state; camera actions change only the observation pose. The model may also call a non-environment tool or stop, neither of which advances the environment.

Camera and robot actions consume the same episode budget. Indexing dispatched camera and robot actions by t , the episode ends at $t_{\text{end}} \leq B_\tau$ when ϕ_τ is satisfied, the model stops, or the budget is exhausted.

3.3 DEMONSTRATION-CONDITIONED RE-GROUNDING

In the standard base protocol, model m inspects multiple distinct frames from an RGB-only video of a successful task execution and writes a structured textual summary $s_{m,\tau}$. It infers constraints such as grasp region, approach, and arm coordination from the visible procedure. Neither the video nor the summary supplies machine-readable actions, robot or object poses, contact labels, semantic phases, or a metric solution for the live scene. The model must re-ground the procedure under a new seed and generate every numerical action parameter (Jang et al., 2022; Jiang et al., 2023; Mandlekar et al., 2023; Wang et al., 2023; Mandlekar et al., 2022).

Held-out transfer variants reuse the base-task summary without a new video. The long-horizon basket task instead provides four atomic-skill videos and no full-task demonstration. Demonstrations

may be omitted only in explicitly named controls. Appendix C.8 gives the summary protocol and controls, and Appendix D describes the frame-inspection gate.

3.4 OBSERVATION, ACTION, AND PRIVILEGE BOUNDARY

Model-visible inputs. The model receives the task instruction, demonstration-derived summary, clean RGB, and filtered proprioception: end-effector pose, gripper state, finger midpoint (GC), and gripper-local axes in world coordinates, for each controlled arm. History retains prior actions and textual feedback, including action validity and trajectory-planning outcomes.

Robot and camera actions. $\mathcal{A}_{\text{robot}}$ is parameterized. The model supplies the magnitude of each world-axis translation (1–100 mm) and gripper-local rotation (1–90 degrees), and issues gripper aperture commands. Dual-arm tasks also support synchronized variants. These numerical targets are generated by the model, not selected from a predefined menu. $\mathcal{A}_{\text{cam}}$ is a bounded discrete set of gripper-centered semantic viewpoints and fixed-step local translation, zoom, yaw, and pitch adjustments. Camera actions test *where* the model chooses to look; robot actions require metric spatial estimation. The camera interface does not control a physical camera arm. Appendices C.5–C.6 give the complete schemas and bounds.

Withheld information. The model does not receive target markers, keypoints, masks, depth, segmentation, object coordinates or poses, contact or task-checker state, task waypoints, camera calibration, offline seed-verification probes, or seed provenance. These signals remain evaluator-only.

3.5 AGENT HARNESS AND FIXED EXECUTION

All models share a harness that maintains interaction history, validates decisions, and dispatches commands. The MLLM is the only model-dependent component, with no learned robot policy or model-specific action head. The fixed executor checks reachability and executes trajectories. The model selects objects, grasp regions, arm roles, waypoints, and corrective actions. The primary direct-control condition uses no learned RGB-D or pre-grasp spatial module. Any such module is reported as a named ablation with its own policy and input manifest. Appendix D gives the implementation details and illustrates the interaction loop in Figure 4.

3.6 TASK CONSTRUCTION AND INSTANCE VALIDATION

The suite is implemented in RoboTwin with a fixed, model-agnostic inverse kinematics and trajectory layer (Mu et al., 2025). Candidate task families must have a visually recoverable spatial bottleneck and support direct control, a deterministic success checker, and observable subtask checkpoints.

Seed construction is deterministic. Each family retains 20 seeds after settling and physical-verification checks. All 280 retained instances pass a complete expert execution from the robot home state to terminal success. The suite is fixed before evaluation and reused unchanged across models and runs. Verification states and actions remain evaluator-only. Appendix C.4 gives the screening protocol, probed counts, randomization ranges, and terminal predicates.

3.7 OUTCOMES AND BEHAVIORAL DIAGNOSTICS

For model m , run r , task τ , and valid seed i , terminal success is the binary outcome of the task-specific RoboTwin checker:

$$y_{m,r,\tau,i} = \mathbf{1}[\phi_{\tau}(x_{t_{\text{end}}}) = 1], \quad t_{\text{end}} \leq B_{\tau}. \quad (2)$$

RoboTwin checks the predicate after each dispatched action. Stops and budget exhaustion without success count as failures, with no post-hoc exclusions. The outcome measures the complete model–interface–executor system.

We complement terminal success with nine trajectory-level behavioral criteria, human-annotated on one complete base-suite run per evaluated condition and grouped into three modules. *Spatial perception and understanding* includes target perception and localization (TL), informative active ex-

ploration (AE), and robot-object spatial relations (SR). *Robot manipulation* includes manipulation-semantics understanding (MS), goal-directed manipulation planning (MP), and fine-grained pre-contact analysis (FG). *Error recovery* includes error detection (ED), online correction (OC), and post-failure adjustment (PF). These criteria help locate failures in the trajectory and complement the environment checker (Liu et al., 2023c; Ren et al., 2023; Duan et al., 2025; Xiong et al., 2025a; Jiang et al., 2025).

Each episode label is the majority vote of three annotators. ED, OC, and PF apply only when an observable error occurs; inapplicable episodes are NA and excluded from their denominators. The other six criteria apply to every annotated base episode. Terminal success remains the primary outcome, and module means are descriptive equal-weight summaries. Appendix C.9 gives the rubric, subtask-progress measure, and annotation examples. Agreement and vote-rule sensitivity appear in Appendix C.9.1.

4 MAIN RESULTS AND ANALYSIS

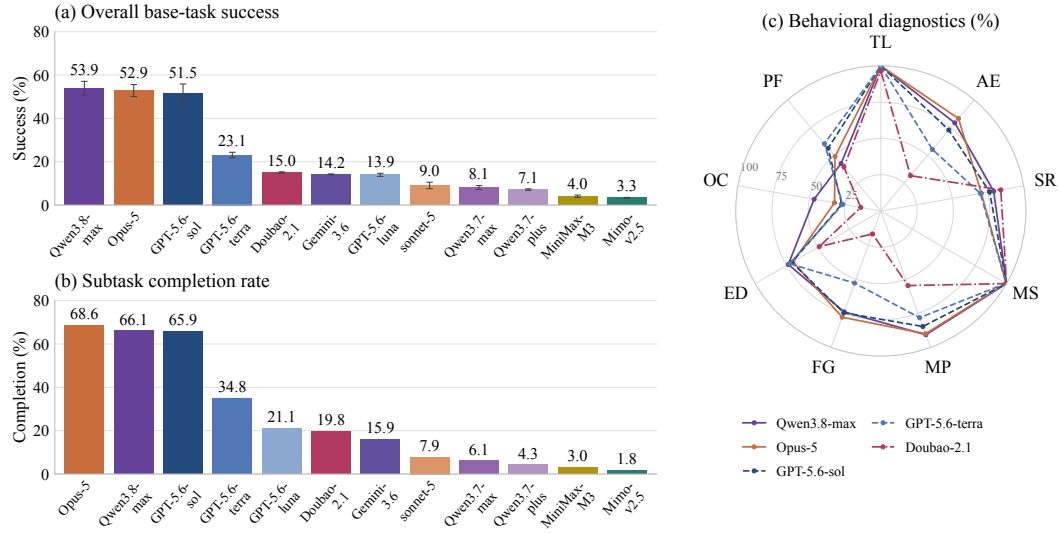


Figure 2: Execution success, subtask progress, and behavioral diagnostics. (a) Mean terminal success and sample standard deviation across three runs on 14 base-task families. (b) Micro-averaged subtask completion from one annotated run per condition. (c) Pooled majority-vote percentages for the five strongest conditions. Panels (a) and (b) include all 12 models; Appendix Table 11 gives their full diagnostic results.

We evaluate 12 models using their own demonstration-derived summaries. Terminal success is a task-family macro-average, reported as mean $\pm$ sample standard deviation across three runs. Diagnostics and subtask progress use one annotated run per model. Appendix A.1 specifies the evaluation conditions and aggregation; complete per-task and arm-specific results appear in Appendix A.

4.1 BASE-TASK PERFORMANCE

Table 3 compares 12 primary model conditions on the 11 single-arm and three dual-arm tasks. Qwen3.8-max has the highest three-run mean at $53.93 \pm 3.17\%$, followed by Opus-5 at $52.86 \pm 2.79\%$ and GPT-5.6-sol at $51.55 \pm 4.31\%$. The top three means lie within 2.38 percentage points, and their run-level ranges overlap. With only three runs, we do not claim a reliable ordering among them. Performance differs across the arm subsets. Qwen3.8-max has the highest single-arm mean (65.61%), and GPT-5.6-sol has the highest dual-arm mean (22.22%).

In the annotated run, the same three conditions lead micro-averaged subtask completion and stay within 2.7 points of one another. Opus-5 reaches 68.57%, Qwen3.8-max 66.07%, and GPT-5.6-sol 65.89%. Opus-5 makes the most intermediate progress, although its terminal success is not

distinguishable from the other two. Progress uses checkpoint-level weighting and is not an upper bound on task-family success.

Performance falls sharply below the top three. GPT-5.6-terra reaches $23.10 \pm 1.25\%$, and every remaining condition is at or below 15.00% . Coordinated dual-arm execution remains difficult for all models. Only the top three conditions exceed a 10% mean, and the best reaches just 22.22%. Appendix Tables 7–9 provide the complete task-by-condition matrix.

Table 3: VA-Bench base-task success (%) for 12 primary model conditions. Entries are mean $\pm$ sample standard deviation across three runs; each run macro-averages 11 single-arm tasks, three dual-arm tasks, or all 14 tasks.

Subset	Qwen3.8 max	Opus 5	GPT-5.6 sol	GPT-5.6 terra	Doubao 2.1	Gemini 3.6
Single-arm	65.61$\pm$3.81	64.09 $\pm$ 3.18	59.55 $\pm$ 5.06	29.24 $\pm$ 1.39	19.09 $\pm$ 0.45	18.03 $\pm$ 0.26
Dual-arm	11.11 $\pm$ 2.55	11.67 $\pm$ 1.67	22.22$\pm$1.92	0.56 $\pm$ 0.96	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00
All 14	53.93$\pm$3.17	52.86 $\pm$ 2.79	51.55 $\pm$ 4.31	23.10 $\pm$ 1.25	15.00 $\pm$ 0.36	14.17 $\pm$ 0.21
Subset	GPT-5.6 luna	sonnet 5	Qwen3.7 max	Qwen3.7 plus	MiniMax M3	Mimo v2.5
Single-arm	17.42 $\pm$ 0.69	11.52 $\pm$ 1.89	10.30 $\pm$ 1.14	9.09 $\pm$ 0.45	5.15 $\pm$ 0.69	4.24 $\pm$ 0.26
Dual-arm	1.11 $\pm$ 0.96	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00
All 14	13.93 $\pm$ 0.71	9.05 $\pm$ 1.49	8.10 $\pm$ 0.90	7.14 $\pm$ 0.36	4.05 $\pm$ 0.55	3.33 $\pm$ 0.21

4.2 INTERACTION AND FAILURE ANALYSIS

Figure 2 shows an uneven diagnostic profile. The top three conditions score 100.0% on target localization (TL) and at least 99.6% on manipulation semantics (MS). Their spatial-relation scores (SR) are lower: 78.9% for Qwen3.8-max, 70.4% for Opus-5, and 76.1% for GPT-5.6-sol. Identifying the target and selecting meaningful motions does not establish accurate robot–object geometry.

Opus-5 has the highest active-exploration (AE) score at 83.2% and the highest fine-grained pre-contact analysis (FG) score at 77.9%. GPT-5.6-sol and Qwen3.8-max follow on FG at 74.6% and 73.9%. GPT-5.6-terra reaches 52.9%, and every remaining condition is below 40%. FG records analysis before contact, not successful operation completion. These trajectory-level scores complement strict terminal success; they do not replace it. Appendix Table 11 reports all 12 conditions.

Recovery performance also varies by criterion. Error detection is comparable across Qwen3.8-max (73.6%), GPT-5.6-terra (72.6%), and Opus-5 (69.9%). Online correction differs sharply: Qwen3.8-max reaches 46.7%, Opus-5 32.4%, and GPT-5.6-terra 26.6%. Gemini-3.6 leads PF at 61.9% despite near-zero OC. High error-detection performance does not imply that a model can correct the deviation online or replan after an explicit failure. The diagnostics record which behaviors occur and where they stop within a trajectory, information that terminal success alone does not provide.

The three annotators agree unanimously on 95.7% of applicable labels (Fleiss $\kappa = 0.942$). Appendix C.9.1 reports per-criterion agreement, sensitivity to the voting rule, and the relationship between diagnostic labels and terminal outcomes.

5 GENERALIZATION AND ABLATION

We first vary the camera observation protocol, then test four forms of robustness beyond the base benchmark: controlled instance transfer, long-horizon skill composition, transfer of a demonstration summary across models, and summary-guided evaluation of locally deployed small models. Seven single-arm families vary geometry, container identity, or layout while reusing the base-task summary. A separate 20-episode track requires five-object placement from four independently demonstrated atomic skills, without a demonstration of the full trajectory. These controlled analyses use their original matched single-run trajectories; the three-run aggregation in Section 4 applies to the primary base-task terminal-success comparison.

5.1 ACTIVE EVIDENCE ACQUISITION VS. PASSIVE MULTI-VIEW INPUT

We compare the canonical active-view protocol with a passive multi-view alternative for four conditions that span the observed performance range: Qwen3.8-max, GPT-5.6-sol, Opus-5, and Doubao-2.1. For each condition, the passive run uses the setting of its first base-evaluation run. We retain the model checkpoint, its own demonstration summary, the same 14 task families with the same 20 verified seeds each, and the same robot-action interface, executor, action budget, terminal checker, and scoring. The context limit and compaction policy are also fixed. Only the camera interface changes, so each model is compared against itself.

The active condition gives the model one current RGB image and semantic or local camera actions, which consume the shared episode budget. The passive condition disables camera actions and supplies five predefined views at every observation point; the model cannot select, reorder, or extend this bundle. The active camera interface can reach the same five canonical viewpoints and additionally supports bounded incremental refinement. The fixed five-view bundle is therefore a strict subset of the viewpoints reachable in active mode.

Each passive request contains only the current five-image bundle. Prior robot actions, planner outcomes, and textual feedback remain in history as in the active condition. This comparison tests model-directed decisions about whether and where to acquire evidence, including local viewpoint refinement, against predefined five-view coverage. Appendix C.7 gives the view definitions and payload order. Table 4 reports pooled results.

Table 4: Controlled single-run active-versus-passive evidence-acquisition comparison for four conditions. Each condition reuses its own base-evaluation summary, the same 20 seeds per family, and the same executor, budget, and checker; only camera control differs. Entries are success count / total episodes followed by success rate, and Δ is passive minus active.

Model	Protocol	Single-arm (11×20)	Dual-arm (3×20)	All 14 (280)
Qwen3.8-max	Active	154/220 (70.00%)	7/60 (11.67%)	161/280 (57.50%)
	Passive	77/220 (35.00%)	1/60 (1.67%)	78/280 (27.86%)
	Δ	−35.00 pp	−10.00 pp	−29.64 pp
GPT-5.6-sol	Active	143/220 (65.00%)	14/60 (23.33%)	157/280 (56.07%)
	Passive	92/220 (41.82%)	3/60 (5.00%)	95/280 (33.93%)
	Δ	−23.18 pp	−18.33 pp	−22.14 pp
Opus-5	Active	144/220 (65.45%)	8/60 (13.33%)	152/280 (54.29%)
	Passive	82/220 (37.27%)	3/60 (5.00%)	85/280 (30.36%)
	Δ	−28.18 pp	−8.33 pp	−23.93 pp
Doubao-2.1	Active	42/220 (19.09%)	0/60 (0.00%)	42/280 (15.00%)
	Passive	15/220 (6.82%)	0/60 (0.00%)	15/280 (5.36%)
	Δ	−12.27 pp	±0.00 pp	−9.64 pp

Removing camera control reduces success by 29.64, 22.14, 23.93, and 9.64 points, respectively. No condition matches its active score with broader passive coverage. The effect spans the tested performance range, including the two strongest conditions, a mid-range condition, and Doubao-2.1. Even its active score of 15.00% falls by nearly two thirds in relative terms. Aggregating at the task-family level, passive observation is worse on 10 of 12 non-tied families for Qwen3.8-max, on all 12 for GPT-5.6-sol and Opus-5, and on 8 of 9 for Doubao-2.1 (two-sided sign tests, $p = 0.039, 0.0005, 0.0005$, and 0.039). The degradation also concentrates on tasks that require task-conditioned evidence. Placement families show sharp declines under passive observation. For Qwen3.8-max, `place_cube_in_bowl` falls from 85.00% to 5.00% and `place_single_cube` from 60.00% to 10.00%. Grasping families degrade far less; these depend on locating a single object that the fixed views already cover. Broader scene coverage does not replace directing observation at the region relevant to the current subgoal. The mechanism behind the gap remains open. It may involve view relevance, cross-view integration, or other factors (Appendix B).

Controlled transfer. Qwen3.8-max transfers best, retaining 67.86% success with a 10.00-point drop. GPT-5.6-sol starts from a comparable matched-base rate but loses 32.14 points. Strong base execution thus does not ensure robust spatial re-grounding. The smaller drops of weaker conditions occur at much lower performance floors. Appendix Table 15 reports all seven families.

Table 5: Controlled transfer and composition. Base and held-out values average seven matched families (20 seeds each); Δ is held-out minus base. Long reports successful object placements out of 100 possible and strict episode success (%) over 20 episodes.

Condition	Matched transfer (%)			Long horizon	
	Base	Held-out	Δ	Objects	Success
Qwen3.8-max	77.86	67.86	-10.00	61/100	0.00
GPT-5.6-sol	80.00	47.86	-32.14	53/100	0.00
GPT-5.6-terra	35.00	27.86	-7.14	5/100	0.00
GPT-5.6-luna	22.86	18.57	-4.29	2/100	0.00

Long-horizon composition. No model completes a strict long-horizon episode. Qwen3.8-max achieves the greatest partial progress with 61/100 successful object placements, followed by GPT-5.6-sol with 53/100. This partial progress is insufficient for complete inventory-conditioned composition: strict task success remains zero.

Trajectory diagnostics. Under controlled transfer, spatial perception and understanding remains at or above 99.3% for all four conditions. Robot manipulation ranges from 54.3% to 86.7%. High spatial-perception module scores coexist with lower manipulation scores and incomplete task execution. Appendix Table 16 reports the three descriptive module means.

5.2 DEMONSTRATION-SUMMARY ABLATION

A matched Qwen3.7-plus comparison holds the evaluated model, task set, and seeds fixed. Replacing its model-specific summary with the Qwen3.8-max summary reduces the 14-task macro-average from 7.14% to 5.00% and leaves dual-arm success at zero. A stronger model’s textual summary is not automatically a better interface for another model. Appendix Table 17 gives the subset breakdown. This comparison covers one reader, so we next test several readers.

Cross-model summary sources. VA-Bench primarily measures how a model understands a task, learns it from a demonstration, and integrates spatial understanding with manipulation. These results combine the quality of the written interface a model produces with its ability to act on that interface. We hold the interface fixed and vary the reader to examine these factors. Shared base summaries come from the three strongest conditions: Qwen3.8-max, GPT-5.6-sol, and Opus-5. We evaluate five models under each summary source on three tasks (`grasp_single_bottle`, `grasp_single_pen`, `place_cube_in_bowl`). For each model, all settings except the summary match its first base-evaluation run. This includes the same 20 verified seeds per task, executor, action budget, and terminal checker. Each cell contains 60 episodes. The *Own* column of Table 6 reports that base run. For the three summary authors, their own-summary cells reproduce it identically, making the shared columns directly comparable to this baseline.

Table 6: Shared demonstration-summary sources. Rates are pooled success (%) over three tasks with 20 seeds each; *Own* is the model’s own model-specific summary in its first base-evaluation run, and Δ is shared minus own. In the *avg* row, rates average all five models; Δ excludes the summary author, whose own-summary cell is its base run by construction.

Evaluated model	Own	On Qwen3.8-max		On GPT-5.6-sol		On Opus-5	
		Rate	Δ	Rate	Δ	Rate	Δ
Qwen3.8-max	86.67	86.67	0.00	85.00	-1.67	88.33	+1.67
Opus-5	85.00	56.67	-28.33	78.33	-6.67	85.00	0.00
GPT-5.6-sol	73.33	21.67	-51.67	73.33	0.00	76.67	+3.33
GPT-5.6-terra	41.67	10.00	-31.67	41.67	0.00	55.00	+13.33
Gemini-3.6	23.33	10.00	-13.33	35.00	+11.67	35.00	+11.67
avg	62.00	37.00	-31.25	62.67	+0.83	68.00	+7.50

The model with the highest task completion does not produce the most transferable summary. Qwen3.8-max leads task completion, but its summary lowers success for all four other models by 31.25 points on average. The largest drop is 51.67 points for GPT-5.6-sol. The Opus-5 summary

improves all four other models by 7.50 points on average. GPT-5.6-sol’s summary is roughly neutral at +0.83.

The content of the summaries suggests a possible explanation. Qwen3.8-max states the fewest constraints and leaves object-specific detail implicit. GPT-5.6-sol and Opus-5 explicitly constrain both fine-grained details and the overall procedure; Opus-5 is the most explicit of the three. A permissive summary may be sufficient for its author but underspecified for the other models. More explicit constraints may allow a summary to transfer with little loss. They may also help the weakest readers most by supplying details that those models omit from their own summaries.

Under the Opus-5 summary, Gemini-3.6 rises from 23.33% to 35.00% and GPT-5.6-terra from 41.67% to 55.00%. The gain is concentrated on the placement task that neither completes reliably on its own (`place_cube_in_bowl`: 0% $\rightarrow$ 20% and 15% $\rightarrow$ 40%). These observations suggest that summary transfer depends in part on how explicitly the constraints are stated, not on the task success of the authoring model. The comparison covers three tasks in a single run. We therefore treat the ordering of the sources as a suggestive trend rather than a precise estimate of transfer effects. Appendix Table 18 gives the per-task breakdown.

5.3 SUMMARY-GUIDED LOCAL SMALL-MODEL EVALUATION

We further evaluate four locally deployed models using summaries sourced from Qwen3.8-max. Qwen3.8-27B and Gemma-4-31B-it obtain 27/280 (9.64%) and 24/280 (8.57%), while the two Qwen3.6 variants reach 3/280 (1.07%) and 2/280 (0.71%). For Gemma-4-31B-it and the Qwen3.6 variants, the corresponding unguided controls obtain zero success. Summary-guided successes remain confined to grasping and bell interaction: no condition completes a placement or dual-arm task. Appendix Tables 19 and 20 provide the aggregate and task-level breakdowns.

6 CONCLUSION

VA-Bench evaluates whether a general-purpose MLLM can close the active observe–reason–act loop through metric robot commands, without target poses, oracle trajectories, or a learned action head. Across 14 task families and three runs, the best macro-average is $53.93 \pm 3.17\%$. Models achieve strong target-localization and manipulation-semantics scores, but spatial-relation diagnostics are less consistent and online correction remains weak.

Strong target-localization scores do not ensure precise, composable, and recoverable physical action. Strict environment success measures whether a model completes the task. The trajectory diagnostics show where models stop using spatial evidence effectively during execution.

In separate matched protocol runs, removing camera control reduces terminal success by 9.64 to 29.64 percentage points across the tested conditions. Qwen3.8-max, for example, falls from 57.50% to 27.86%. GPT-5.6-sol, Opus-5, and Doubao-2.1 show the same direction. Broader passive coverage does not substitute for task-conditioned evidence acquisition.

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A COMPLETE EVALUATION RESULTS

This appendix reports the disaggregated values underlying the compact tables in the main paper. All tables remain in the normal reading orientation. The per-task base-benchmark matrix is split by model group, and the nine behavioral diagnostics are split by module. Model columns and rows follow descending 14-task macro-average; bold blue cells mark the best score within each directly comparable task row or metric column, including ties.

A.1 BASE-EVALUATION SETTINGS AND AGGREGATION

Conditions and validity. The primary comparison evaluates 12 models, each using a summary it produced from the task’s RGB-only expert video. Two shared-summary controls reuse another model’s summary and are reported separately in Section 5: one tests summary-source transfer and the other supports the local-model comparison. Their outcomes are not pooled with the primary conditions.

Following Section 3.6, each independent run uses the same 20 retained seeds for each of the 11 single-arm and three dual-arm families. Three runs give 60 outcomes per task and 840 run–seed outcomes per primary condition. Offline verification signals remain evaluator-only, and terminal success is determined solely by the environment checker. All models share the observation, action, budget, and executor contract in Section 3; harness details appear in Appendix D.

Terminal success. Let $y_{m,r,\tau,i} \in \{0, 1\}$ be the checker outcome for model m , run r , task τ , and seed i . We average over seeds within a task and then macro-average over task families:

$$S_{m,r,\tau} = \frac{1}{20} \sum_{i=1}^{20} y_{m,r,\tau,i}, \quad S_{m,r}^{\text{all}} = \frac{1}{14} \sum_{\tau=1}^{14} S_{m,r,\tau}. \quad (3)$$

We report the mean and sample standard deviation of the three run-level macro-averages. Single- and dual-arm scores apply the same procedure over 11 and three families, respectively. Every family has equal weight, and the error bars measure run-to-run variation, not binomial uncertainty over seeds.

Annotated run and partial progress. Diagnostics and subtask progress use one complete base run per primary condition, totaling 3,360 episodes. Eleven conditions use their first run; sonnet-5 uses its second. Each diagnostic pools applicable episode-level majority-vote labels across the single- and dual-arm subsets. Figure 2 plots these percentages directly, without averaging individual annotator votes or applying min–max normalization.

Subtask completion is a micro-average over scored checkpoint instances, with 420 single-arm and 140 dual-arm checkpoints per model. Because its weighting unit is a checkpoint rather than a task family, it measures intermediate progress but is not a relaxed upper bound on terminal success. Appendix C.9 defines observable checkpoint evidence and diagnostic applicability. Appendix C.9.1 reports agreement and sensitivity analyses.

A.2 BASE-TASK SUCCESS RATES

Tables 7–9 report the three-run mean and sample standard deviation for each of the 14 base tasks and 12 primary model conditions. Each run evaluates the same 20 retained seeds meeting the acceptance criteria in Appendix C.4; the final row aggregates each run over all 14 tasks before computing summary statistics. Steps is the maximum interaction horizon. Doubao-2.1 and Gemini-3.6 abbreviate `doubao-seed-2.1-turbo` and `gemini-3.6-flash`, respectively.

Table 7: Base-task success rates (% , mean $\pm$ sample SD), model group 1 of 3.

Task	Steps	Qwen3.8 max	Opus 5	GPT-5.6 sol	GPT-5.6 terra
<i>Single-arm tasks</i>					
beat_block_hammer_right	100	3.33 $\pm$ 2.89	8.33 $\pm$ 2.89	15.00$\pm$5.00	6.67 $\pm$ 2.89
click_bell_right	50	73.33$\pm$5.77	68.33 $\pm$ 14.43	65.00 $\pm$ 15.00	1.67 $\pm$ 2.89
grasp_pen_leaning_cube	50	61.67$\pm$2.89	35.00 $\pm$ 5.00	45.00 $\pm$ 18.03	25.00 $\pm$ 5.00
grasp_single_bottle	50	91.67$\pm$2.89	75.00 $\pm$ 0.00	76.67 $\pm$ 11.55	51.67 $\pm$ 5.77
grasp_single_bottle_upright	50	65.00 $\pm$ 5.00	78.33 $\pm$ 16.07	70.00 $\pm$ 10.00	68.33 $\pm$ 2.89
grasp_single_cube	50	83.33$\pm$10.41	83.33$\pm$11.55	76.67 $\pm$ 17.56	31.67 $\pm$ 5.77
grasp_single_pen	50	91.67$\pm$10.41	63.33 $\pm$ 15.28	28.33 $\pm$ 16.07	48.33 $\pm$ 7.64
place_cube_in_bowl	50	81.67 $\pm$ 2.89	88.33$\pm$10.41	85.00 $\pm$ 10.00	18.33 $\pm$ 2.89
place_cube_on_cube	200	86.67 $\pm$ 7.64	78.33 $\pm$ 2.89	88.33$\pm$12.58	46.67 $\pm$ 2.89
place_single_bottle_upright	100	40.00$\pm$5.00	38.33 $\pm$ 2.89	13.33 $\pm$ 2.89	6.67 $\pm$ 5.77
place_single_cube	100	43.33 $\pm$ 14.43	88.33 $\pm$ 7.64	91.67$\pm$7.64	16.67 $\pm$ 2.89
<i>Dual-arm tasks</i>					
place_shoe	100	33.33 $\pm$ 7.64	35.00 $\pm$ 5.00	66.67$\pm$5.77	0.00 $\pm$ 0.00
handover_horizontal_block	100	0.00$\pm$0.00	0.00$\pm$0.00	0.00$\pm$0.00	0.00$\pm$0.00
lift_pot	50	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	1.67$\pm$2.89
<i>Macro-average</i>		53.93$\pm$3.17	52.86 $\pm$ 2.79	51.55 $\pm$ 4.31	23.10 $\pm$ 1.25

Table 8: Base-task success rates (% , mean $\pm$ sample SD), model group 2 of 3.

Task	Steps	Doubao 2.1	Gemini 3.6	GPT-5.6 luna	sonnet 5
<i>Single-arm tasks</i>					
beat_block_hammer_right	100	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00
click_bell_right	50	6.67 $\pm$ 5.77	13.33 $\pm$ 2.89	0.00 $\pm$ 0.00	11.67 $\pm$ 2.89
grasp_pen_leaning_cube	50	6.67 $\pm$ 2.89	10.00 $\pm$ 5.00	18.33 $\pm$ 2.89	5.00 $\pm$ 8.66
grasp_single_bottle	50	86.67 $\pm$ 2.89	38.33 $\pm$ 2.89	3.33 $\pm$ 5.77	33.33 $\pm$ 5.77
grasp_single_bottle_upright	50	48.33 $\pm$ 2.89	78.33 $\pm$ 2.89	86.67$\pm$5.77	40.00 $\pm$ 10.00
grasp_single_cube	50	10.00 $\pm$ 5.00	8.33 $\pm$ 2.89	15.00 $\pm$ 5.00	16.67 $\pm$ 2.89
grasp_single_pen	50	26.67 $\pm$ 10.41	23.33 $\pm$ 7.64	8.33 $\pm$ 2.89	1.67 $\pm$ 2.89
place_cube_in_bowl	50	6.67 $\pm$ 7.64	5.00 $\pm$ 5.00	6.67 $\pm$ 7.64	11.67 $\pm$ 2.89
place_cube_on_cube	200	8.33 $\pm$ 2.89	10.00 $\pm$ 5.00	13.33 $\pm$ 2.89	0.00 $\pm$ 0.00
place_single_bottle_upright	100	6.67 $\pm$ 2.89	11.67 $\pm$ 2.89	26.67 $\pm$ 7.64	0.00 $\pm$ 0.00
place_single_cube	100	3.33 $\pm$ 2.89	0.00 $\pm$ 0.00	13.33 $\pm$ 2.89	6.67 $\pm$ 2.89
<i>Dual-arm tasks</i>					
place_shoe	100	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	3.33 $\pm$ 2.89	0.00 $\pm$ 0.00
handover_horizontal_block	100	0.00$\pm$0.00	0.00$\pm$0.00	0.00$\pm$0.00	0.00$\pm$0.00
lift_pot	50	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00
<i>Macro-average</i>		15.00 $\pm$ 0.36	14.17 $\pm$ 0.21	13.93 $\pm$ 0.71	9.05 $\pm$ 1.49

Table 9: Base-task success rates (% , mean $\pm$ sample SD), model group 3 of 3.

Task	Steps	Qwen3.7 max	Qwen3.7 plus	MiniMax M3	Mimo v2.5
<i>Single-arm tasks</i>					
beat_block_hammer_right	100	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00
click_bell_right	50	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00
grasp_pen_leaning_cube	50	1.67 $\pm$ 2.89	3.33 $\pm$ 2.89	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00
grasp_single_bottle	50	1.67 $\pm$ 2.89	18.33 $\pm$ 2.89	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00
grasp_single_bottle_upright	50	76.67 $\pm$ 7.64	76.67 $\pm$ 2.89	46.67 $\pm$ 2.89	41.67 $\pm$ 10.41
grasp_single_cube	50	20.00 $\pm$ 10.00	0.00 $\pm$ 0.00	10.00 $\pm$ 5.00	5.00 $\pm$ 8.66
grasp_single_pen	50	13.33 $\pm$ 2.89	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00
place_cube_in_bowl	50	0.00 $\pm$ 0.00	1.67 $\pm$ 2.89	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00
place_cube_on_cube	200	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00
place_single_bottle_upright	100	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00
place_single_cube	100	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00
<i>Dual-arm tasks</i>					
place_shoe	100	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00
handover_horizontal_block	100	0.00$\pm$0.00	0.00$\pm$0.00	0.00$\pm$0.00	0.00$\pm$0.00
lift_pot	50	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00
<i>Macro-average</i>		8.10 $\pm$ 0.90	7.14 $\pm$ 0.36	4.05 $\pm$ 0.55	3.33 $\pm$ 0.21

Table 10: Per-task successes out of 20 seeds for the controlled active evidence acquisition versus passive five-view observation comparison, first base-evaluation run. Each condition reuses its own demonstration summary and the same seeds in both protocols; robot execution and terminal success/failure accounting are identical. The active protocol permits camera actions, whereas the passive protocol supplies five predefined views without camera control. A/P denotes active/passive.

Task	Qwen3.8-max		GPT-5.6-sol		Opus-5		Doubao-2.1	
	A	P	A	P	A	P	A	P
<i>Single-arm tasks</i>								
click_bell_right	16	19	16	14	12	10	0	3
grasp_pen_leaning_cube	13	11	5	2	8	6	1	0
grasp_single_bottle	19	12	18	11	15	13	17	7
grasp_single_bottle_upright	14	8	16	13	12	7	10	2
grasp_single_cube	19	7	19	15	18	11	3	0
grasp_single_pen	16	11	7	6	16	13	7	3
place_cube_in_bowl	17	1	19	17	20	8	0	0
place_cube_on_cube	19	5	20	14	16	6	2	0
place_single_bottle_upright	9	0	3	0	8	2	1	0
place_single_cube	12	2	17	0	18	6	1	0
beat_block_hammer_right	0	1	3	0	1	0	0	0
<i>Single-arm total (/220)</i>	154	77	143	92	144	82	42	15
<i>Dual-arm tasks</i>								
place_shoe	7	1	14	3	8	3	0	0
handover_horizontal_block	0	0	0	0	0	0	0	0
lift_pot	0	0	0	0	0	0	0	0
<i>Dual-arm total (/60)</i>	7	1	14	3	8	3	0	0
<i>All-task total (/280)</i>	161	78	157	95	152	85	42	15

A.3 BASE-TASK BEHAVIORAL DIAGNOSTICS

Tables 12–14 give the single- and dual-arm components behind the pooled scores in Table 11. Each pooled value combines valid counts from the two subsets. All four tables report the same twelve primary conditions, each with one fully annotated run. For eleven conditions the annotated run is the first run; for sonnet-5 it is the second (13.64% single-arm success). Every component-table entry is a percentage. TL: target perception and localization; AE: informative active exploration; SR: robot–object spatial relations; MS: manipulation-semantics understanding; MP: goal-directed manipulation planning; FG: fine-grained pre-contact analysis, scored over every reported episode; ED: error detection; OC: online correction; PF: post-failure adjustment. Annotators marked an assessable ambiguity in every base-task episode, so AE is applicable to all 280 annotated episodes per condition in this track; the error-recovery criteria retain their majority-voted applicability gate. For ED, OC, and PF, an episode with no observable error is recorded as NA and excluded from the denominator. When an error occurs, all three criteria are applicable and receive an explicit positive or negative judgment.

Table 11: Pooled base-task trajectory-level behavioral diagnostics (%) for all twelve primary conditions. Scores combine applicable single- and dual-arm episodes; error-free episodes are excluded from ED, OC, and PF.

Condition	TL	AE	SR	MS	MP	FG	ED	OC	PF
Qwen3.8-max	100.0	79.3	78.9	100.0	90.7	73.9	73.6	46.7	42.6
Opus-5	100.0	83.2	70.4	99.6	90.0	77.9	69.9	32.4	49.1
GPT-5.6-sol	100.0	72.9	76.1	100.0	84.6	74.6	70.8	27.3	56.5
GPT-5.6-terra	100.0	55.4	69.6	99.6	78.2	52.9	72.6	26.6	60.5
Doubao-2.1	96.4	31.8	83.9	100.0	54.6	16.8	49.0	14.1	39.5
Gemini-3.6	100.0	31.4	85.4	100.0	73.2	8.9	63.8	1.9	61.9
GPT-5.6-luna	100.0	40.0	71.8	99.3	61.1	38.9	49.2	13.7	37.4
sonnet-5	99.3	21.8	80.7	100.0	31.8	32.1	10.6	1.2	9.8
Qwen3.7-max	96.4	13.6	81.4	97.5	35.7	2.5	32.6	0.4	32.2
Qwen3.7-plus	97.9	10.7	82.9	94.6	36.8	2.9	17.7	1.1	17.0
MiniMax-M3	95.7	6.1	93.6	95.4	21.4	1.8	22.6	1.4	21.1
Mimo-v2.5	83.2	5.4	71.8	98.9	19.6	2.1	25.8	0.0	26.2

Table 12: Base-task spatial perception and understanding diagnostics (%) for all twelve primary conditions.

Condition	Single-arm			Dual-arm		
	TL	AE	SR	TL	AE	SR
Qwen3.8-max	100.0	85.0	89.1	100.0	58.3	41.7
Opus-5	100.0	90.5	71.4	100.0	56.7	66.7
GPT-5.6-sol	100.0	81.8	81.8	100.0	40.0	55.0
GPT-5.6-terra	100.0	64.1	70.0	100.0	23.3	68.3
Doubao-2.1	95.5	34.1	85.5	100.0	23.3	78.3
Gemini-3.6	100.0	36.8	89.1	100.0	11.7	71.7
GPT-5.6-luna	100.0	47.7	69.5	100.0	11.7	80.0
sonnet-5	99.5	25.5	76.8	98.3	8.3	95.0
Qwen3.7-max	99.1	16.4	83.6	86.7	3.3	73.3
Qwen3.7-plus	99.1	13.2	84.1	93.3	1.7	78.3
MiniMax-M3	96.4	7.3	94.1	93.3	1.7	91.7
Mimo-v2.5	82.7	5.5	69.5	85.0	5.0	80.0

Table 13: Base-task robot manipulation diagnostics (%) for all twelve primary conditions.

Condition	Single-arm			Dual-arm		
	MS	MP	FG	MS	MP	FG
Qwen3.8-max	100.0	94.1	75.5	100.0	78.3	68.3
Opus-5	99.5	94.5	86.4	100.0	73.3	46.7
GPT-5.6-sol	100.0	90.0	76.4	100.0	65.0	68.3
GPT-5.6-terra	99.5	86.4	58.2	100.0	48.3	33.3
Doubao-2.1	100.0	61.4	15.0	100.0	30.0	23.3
Gemini-3.6	100.0	78.2	8.6	100.0	55.0	10.0
GPT-5.6-luna	99.1	63.2	45.0	100.0	53.3	16.7
sonnet-5	100.0	33.6	40.0	100.0	25.0	3.3
Qwen3.7-max	96.8	43.2	3.2	100.0	8.3	0.0
Qwen3.7-plus	93.2	43.6	3.6	100.0	11.7	0.0
MiniMax-M3	94.1	21.8	2.3	100.0	20.0	0.0
Mimo-v2.5	98.6	25.0	2.7	100.0	0.0	0.0

Table 14: Base-task error recovery diagnostics (%) for all twelve primary conditions. Error-free episodes are NA and excluded before computing each percentage.

Condition	Single-arm			Dual-arm		
	ED	OC	PF	ED	OC	PF
Qwen3.8-max	81.7	54.2	43.0	52.7	27.3	41.8
Opus-5	79.4	36.9	55.6	42.9	19.6	30.4
GPT-5.6-sol	76.9	28.8	60.6	53.6	23.2	44.6
GPT-5.6-terra	76.7	34.4	60.8	59.3	1.7	59.3
Doubao-2.1	53.2	18.2	40.9	35.0	0.0	35.0
Gemini-3.6	65.5	2.5	63.0	58.3	0.0	58.3
GPT-5.6-luna	49.0	17.3	34.2	50.0	1.7	48.3
sonnet-5	9.3	1.5	8.2	15.0	0.0	15.0
Qwen3.7-max	29.1	0.5	28.6	45.0	0.0	45.0
Qwen3.7-plus	12.4	1.4	11.5	36.7	0.0	36.7
MiniMax-M3	22.8	1.8	21.0	21.7	0.0	21.7
Mimo-v2.5	32.4	0.0	32.9	1.7	0.0	1.7

A.4 GENERALIZATION AND LONG-HORIZON DIAGNOSTICS

Table 15: Single-arm controlled generalization success rates (%). Each entry uses 20 held-out seeds and reuses the corresponding base-task demonstration summary.

Task family	Qwen3.8 max	GPT-5.6 sol	GPT-5.6 terra	GPT-5.6 luna
grasp_single_bottle	85	80	60	20
grasp_single_bottle_upright	40	35	50	55
grasp_single_cube	80	50	40	15
place_cube_in_bowl	70	40	5	20
place_cube_on_cube	80	65	5	10
place_single_bottle_upright	55	20	15	10
place_single_cube	65	45	20	0
<i>Macro-average</i>	67.86	47.86	27.86	18.57

Table 16 summarizes the same held-out trajectories at the module level. These equal-weight means are descriptive profiles rather than an overall ranking.

Table 16: Overall module-level diagnostic overview (%) under controlled held-out transfer. SPU, RM, and ER equally average TL/AE/SR, MS/MP/FG, and ED/OC/PF, respectively.

Condition	SPU	RM	ER
Qwen3.8-max	99.8	86.7	59.8
GPT-5.6-sol	99.3	76.4	59.9
GPT-5.6-terra	100.0	66.2	60.2
GPT-5.6-luna	100.0	54.3	34.4

A.5 DEMONSTRATION-SUMMARY SOURCE ABLATION

Table 17 gives the subset breakdown for the matched Qwen3.7-plus comparison summarized in Section 5.

Table 17: Effect of demonstration-summary source for Qwen3.7-plus. Columns are macro-averages over 11 single-arm tasks, three dual-arm tasks, and all 14 tasks, respectively.

Summary source	Single-arm	Dual-arm	All 14
Qwen3.7-plus (model-specific)	9.09	0.00	7.14
Qwen3.8-max (shared)	6.36	0.00	5.00
Shared — model-specific	−2.73	0.00	−2.14

Table 18 gives the per-task breakdown of the cross-model shared-summary comparison summarized in Section 5.2. Each entry is the success rate over the same 20 verified seeds used in the corresponding first base-evaluation run; only the demonstration summary is replaced. The own-summary block of each summary author reproduces its base run exactly.

A.6 LOCALLY DEPLOYED SMALL-MODEL RESULTS

Table 20 gives the task-level results summarized in Table 19. Each task contains 20 trajectories. Gemma-4-31B-it and the two Qwen3.6 variants use the Qwen3.8-max demonstration summary, while Qwen3.8-27B uses a mixed-summary condition whose summary source is also Qwen3.8-max. Without summary guidance, Gemma-4-31B-it and the two Qwen3.6 variants obtain 0.00% on every task.

Table 18: Per-task success rate (%) under shared demonstration summaries, 20 seeds per cell. Tasks are B = grasp_single_bottle, P = grasp_single_pen, C = place_cube_in_bowl. *Own* is the model’s own model-specific summary in its first base-evaluation run.

Evaluated model	Own			On Qwen3.8-max			On GPT-5.6-sol			On Opus-5		
	B	P	C	B	P	C	B	P	C	B	P	C
Qwen3.8-max	95	80	85	95	80	85	90	80	85	95	85	85
Opus-5	75	80	100	55	40	75	70	75	90	75	80	100
GPT-5.6-sol	90	35	95	30	15	20	90	35	95	90	45	95
GPT-5.6-terra	55	55	15	20	5	5	65	30	30	60	65	40
Gemini-3.6	40	30	0	15	15	0	60	30	15	50	35	20
Mean	71.00	56.00	59.00	43.00	31.00	37.00	75.00	50.00	63.00	74.00	62.00	68.00

Table 19: Summary-guided success for locally deployed models. Entries are success count / total trajectories followed by the success rate. The single-arm, dual-arm, and overall denominators are 220, 60, and 280, respectively.

Condition	Summary	Single-arm	Dual-arm	All 14
Qwen3.8-27B	Qwen3.8-max	27/220 (12.27%)	0/60 (0.00%)	27/280 (9.64%)
Gemma-4-31B-it	Qwen3.8-max	24/220 (10.91%)	0/60 (0.00%)	24/280 (8.57%)
Qwen3.6-27B	Qwen3.8-max	3/220 (1.36%)	0/60 (0.00%)	3/280 (1.07%)
Qwen3.6-35B-A3B	Qwen3.8-max	2/220 (0.91%)	0/60 (0.00%)	2/280 (0.71%)

Table 20: Per-task success rates (%) for summary-guided locally deployed models. Each task contains 20 trajectories. Bold blue marks the highest nonzero result in each row.

Task	Gemma-4 31B-it	Qwen3.6 27B	Qwen3.6 35B-A3B	Qwen3.8 27B
<i>Single-arm tasks</i>				
click_bell_right	10.00	0.00	0.00	10.00
grasp_pen_leaning_cube	5.00	0.00	0.00	10.00
grasp_single_bottle	30.00	0.00	5.00	15.00
grasp_single_bottle_upright	55.00	15.00	5.00	85.00
grasp_single_cube	5.00	0.00	0.00	10.00
grasp_single_pen	15.00	0.00	0.00	5.00
place_cube_in_bowl	0.00	0.00	0.00	0.00
place_cube_on_cube	0.00	0.00	0.00	0.00
place_single_bottle_upright	0.00	0.00	0.00	0.00
place_single_cube	0.00	0.00	0.00	0.00
beat_block_hammer_right	0.00	0.00	0.00	0.00
<i>Dual-arm tasks</i>				
place_shoe	0.00	0.00	0.00	0.00
handover_horizontal_block	0.00	0.00	0.00	0.00
lift_pot	0.00	0.00	0.00	0.00
<i>Single-arm macro-average</i>	10.91	1.36	0.91	12.27
<i>Dual-arm macro-average</i>	0.00	0.00	0.00	0.00
<i>All-14 macro-average</i>	8.57	1.07	0.71	9.64

B SCOPE AND LIMITATIONS

B.1 WHAT THE DIAGNOSTICS AND PROTOCOL COMPARISONS IDENTIFY

The nine criteria are observational: they record where in a trajectory the behavior visibly breaks, and they support the localization claims of Section 4.2. Attributing a breakpoint to perception, planning, numerical grounding, or execution would require interventions on those components, which this benchmark does not perform. The same applies to the protocol comparisons. The active–passive result of Section 5.1 shows, for the four conditions tested, that removing camera control costs success even when passive input supplies all five canonical views at each observation. These fixed views form a strict subset of the viewpoints reachable in active mode, which additionally allows bounded incremental refinement. The comparison leaves open whether the cost comes from view relevance, cross-view integration, or the longer visual context, and it says nothing about whether multiple simultaneous views are harmful in other settings. Similarly, the cross-model summary comparison

of Section 5.2 varies the summary source over three tasks in a single run. It therefore shows that the source matters and that the ordering of the three sources is not the ordering of their authors’ task success, but the summaries differ in style as well as in constraint explicitness, so it does not identify which property carries the effect.

B.2 GENERALIZATION SCOPE

The held-out variants are finite and controlled; they do not establish open-set generalization across arbitrary objects, robots, cameras, or real-world calibration. The long-horizon result covers one compositional family, and the local-model comparison covers four deployments under summary-guided conditions. Broader causal claims require additional object families and deployment settings.

B.3 LIMITATIONS: DEMONSTRATION-SUMMARY CONDITIONING

The primary results are end-to-end condition comparisons: each model distills its own demonstration summary and then uses it during control. Outcomes therefore combine summary quality, live-scene re-grounding, and action generation rather than measuring summary-independent online control. The shared-summary comparison of Section 5.2 controls the summary source across five models, but only over three tasks in a single run, and the matched Qwen3.7-plus study changes it for one model over the full task set. Fully separating these factors requires no-summary and expert-written summary controls across multiple models on the complete suite, which would also distinguish constraint explicitness from other stylistic differences between summaries.

B.4 LIMITATIONS: MODULAR VLM–VLA CONTROL

The present benchmark evaluates a direct-control setting in which the VLM produces metric robot and camera commands through a fixed, model-agnostic harness. We did not evaluate a modular architecture in which a VLM performs high-level reasoning and instruction generation while a separately trained VLA translates those instructions into low-level robot actions (Zitkovich et al., 2023; O’Neill et al., 2024; Ghosh et al., 2024; Kim et al., 2025; Li et al., 2024). Consequently, the reported results do not measure the benefits or failure modes introduced by a VLM–VLA interface, a learned action policy, or a division of responsibility between high-level planning and low-level control (Shridhar et al., 2022; 2023; Brohan et al., 2023; Jiang et al., 2023; Goyal et al., 2023; Chi et al., 2023). Settling how such a decomposition compares would require a common VLA controller, an explicit instruction interface, and matched training and calibration protocols.

C BENCHMARK SPECIFICATION DETAILS

VA-Bench evaluates whether a general-purpose MLLM can turn partial visual evidence into spatially grounded robot behavior. In the canonical protocol, each episode couples active observation, cross-view reasoning, numerical end-effector control, and physics-simulated task completion. The benchmark does not ask the model to select an answer or a reference trajectory. Instead, the model must determine what evidence is missing, acquire an informative view, infer object–robot geometry, and instantiate that inference as a sequence of metric actions. The camera interface deliberately isolates task-conditioned viewpoint selection from physical camera-arm control, while the trajectory rubric reports behavioral evidence rather than direct measurements of latent cognitive states. The matched passive five-view comparison instead supplies five predefined RGB views in each observation and exposes no camera-action command. Robot actions, seeds, execution, scoring, and context-budget settings remain the same. Each passive API request contains only the current five-image bundle; prior robot actions, planner outcomes, and textual feedback remain in the retained history, subject to the same context limit and compaction policy as the active condition. Thus the comparison isolates the evidence-acquisition protocol rather than robot control or history retention.

C.1 INTERACTIVE TASK FORMULATION

The following formulation describes the canonical active-view protocol. An evaluation instance is

$$z = (\tau, \xi, g_{\tau, \xi}, D_{\tau}, B_{\tau}, \phi_{\tau}), \quad (4)$$

defined by a task family τ , a scene seed ξ , a natural-language goal $g_{\tau,\xi}$, an optional observation-only demonstration D_τ , an episode budget B_τ , and a physics-based task-success predicate ϕ_τ . Before test-time control, the designated summary model converts D_τ into a condition-specific textual summary $s_\tau^{(c)}$; the primary condition uses the evaluated model itself, whereas shared-summary controls use the named source model. The simulator maintains an unobserved physical state x_t . At step t , the model receives

$$o_t = (I_t, p_t, h_t, f_{t-1}), \quad (5)$$

where I_t is the current active-camera RGB image, p_t is filtered robot proprioception, h_t is the interaction history, and f_{t-1} contains the previous action-validity and trajectory-planning outcomes. At each decision, the MLLM may call a non-environment inspection or demonstration tool, stop, or choose one camera or robot action. Let $\mathcal{A}_{\text{robot}}$ and $\mathcal{A}_{\text{cam}}$ denote the robot and camera action sets, respectively:

$$a_t \sim M_\theta(g_{\tau,\xi}, s_\tau^{(c)}, o_{\leq t}, a_{< t}), \quad \begin{cases} x_{t+1} = \mathcal{T}_\tau(x_t, a_t), & a_t \in \mathcal{A}_{\text{robot}}, \\ x_{t+1} = x_t, & a_t \in \mathcal{A}_{\text{cam}}. \end{cases} \quad (6)$$

and observes the resulting scene before choosing again. Robot actions update the physical state, whereas camera actions update only the observation pose. Tool calls and stop decisions do not advance the environment; camera and robot actions consume the same task-dependent budget. Indexing dispatched environment actions by t , an episode ends at $t_{\text{end}} \leq B_\tau$ when $\phi_\tau(x_{t_{\text{end}}})$ becomes true, the model stops, or the budget is exhausted. The primary outcome is therefore binary physics-simulated task success, rather than agreement with an annotated answer or action sequence.

For the passive five-view comparison, the single actively controlled image I_t is replaced by a fixed-order bundle of five predefined RGB views, and $\mathcal{A}_{\text{cam}}$ is removed from the model’s available action set. The passive API sends only the current bundle; prior robot actions, planner outcomes, and textual feedback remain in a history governed by the same context limit and compaction policy as the canonical evaluation. The other observation fields, robot action space, action timing, seeds, executor, budget, success predicate, and scoring are identical. The model cannot select, reorder, or request an additional view in this condition.

Every RGB frame in the observation payload is clean: model-facing images contain no drawn keypoints, coordinate axes, masks, or target markers. Proprioception contains the controlled end-effector position and orientation, gripper open/closed state, the physical finger midpoint (GC), and the current gripper-local axes expressed in the world frame. In dual-arm episodes, these quantities are provided separately for both arms. The model retains a persistent history of prior robot actions, planner outcomes, and textual feedback, bounded by the same context limit and compaction policy in both conditions. The passive API supplies only the current five-view bundle and cannot obtain additional views. The model is not given depth, point clouds, segmentation, object coordinates or poses, contact state, oracle grasp points, task waypoints, or numerical camera calibration. Consequently, the numerical robot state anchors a metric reference frame by design; the benchmark does not test recovery of the robot coordinate system itself. The task-relevant object geometry and its relation to that frame must still be inferred from RGB and interaction.

C.2 TASK-LEVEL SPATIAL DEMANDS

At the task-design level, VA-Bench spans seven spatial–action demands that become jointly observable only when perception is coupled to action.

Active evidence acquisition. The initial observation is not assumed to expose every task-relevant dimension. The model must decide whether uncertainty concerns world- XY position, height, depth, orientation, clearance, or occlusion, and select a viewpoint that is diagnostic for that uncertainty. Because observation draws on the shared action budget, the model must also stop looking and act once the accumulated evidence is sufficient.

Cross-view consistency and online state tracking. Camera motion changes image coordinates but not the world-frame robot controls. The model must reconcile observations across viewpoints,

retain object identity and spatial relations, and invalidate stale evidence after either the camera, gripper, or object moves. After execution, it must interpret the new image and planner outcome relative to its previous action rather than restarting from an independent frame.

Metric object–robot grounding. The model must translate qualitative visual relations into signed world-axis translations and numerical distances. This requires relating an object seen in pixels to GC and end-effector coordinates, distinguishing camera-frame appearance from world-frame motion, and estimating whether the remaining lateral, vertical, and depth offsets are within grasp or placement tolerance.

Orientation, affordance, and contact geometry. Successful interaction depends on more than object centering. The model must infer long axes, surfaces, support relations, graspable regions, functional parts, and approach corridors. Examples include orienting the gripper closing axis across a bottle or pen, grasping a hammer by its handle while aligning the head with a block, pressing the top center of a bell, and preserving an upright container during placement.

Goal-directed metric action sequencing and outcome-conditioned revision. The model must compose approach, grasp, lift, transport, insertion, release, and retreat phases using only incremental metric commands. Low-level inverse kinematics, collision checking, and trajectory generation remain delegated to the fixed execution layer. Re-observation after each action enables online correction when a proposed translation is rejected, an object moves unexpectedly, a grasp misses, or a placement is not yet stable. The trajectory-level diagnostics measure the resulting recovery behavior at the level of observable actions (Appendix B).

Embodiment and bimanual coordination. Tasks with two available arms require reasoning about reachability, arm–object assignment, shared-workspace clearance, and temporal dependencies between roles. Handover tasks require complementary grasp regions and ordered release, whereas pot lifting requires mirrored handle geometry and synchronized motion.

Demonstration-conditioned spatial re-grounding. The demonstration communicates visible task procedure, not the live scene’s metric solution. The model must extract transferable constraints and re-ground them under a new seed, object pose, arm assignment, or held-out geometry. This separates recognizing a demonstrated procedure from executing it in the current spatial configuration.

These seven demands are design dimensions that describe what the task suite requires; they are not seven additional scores or independently identified latent variables. Section C.9 instead operationalizes executed behavior through nine scored trajectory-level diagnostics in three modules. The two levels are intentionally not one-to-one: for example, bimanual coordination is a task condition and demonstration-conditioned re-grounding is evaluated through the transfer protocol. End-to-end outcomes also depend on numerical action generation, the fixed motion planner, and simulated dynamics.

C.3 TASK-FAMILY COVERAGE

VA-Bench comprises 14 base task families: eleven expose a single controllable arm, while three expose two arms and require the model to reason explicitly about their use. In `place_shoe`, the model must select and control the arm on the object’s reachable side; `handover_horizontal_block` and `lift_pot` additionally require coordinated use of both arms. We therefore classify all three as dual-arm tasks, regardless of whether both arms move simultaneously.

In terms of task demands, the single-arm suite tests target pose and long-axis inference, object–robot alignment, grasp-region and approach-direction selection, and placement constraints such as containment, stacking, pose preservation, and release height. Bell pressing and hammering further test functional-part localization, contact direction, and tool geometry. The dual-arm suite requires reachability-aware arm assignment, ordered handover, or synchronized motion. The separately evaluated five-object task tests long-horizon state maintenance, repeated spatial re-grounding, and composition of atomic skills. These demands describe task coverage rather than additional independent scores.

Instances and controlled splits. A scene seed selects task-specific object positions, orientations, and, where configured, asset model IDs. Resulting layouts or arm assignments follow from these task fields rather than from a separate global randomizer. The base benchmark uses 20 retained seeds per task under the verification protocol in Appendix C.4. The primary terminal-success evaluation repeats this fixed suite in three independent runs per model; probe-only simulator state and actions are never exposed to the evaluated model. Seven base families additionally have paired held-out variants: horizontal and upright bottle grasping, cube grasping, cube-in-bowl placement, cube stacking, upright-bottle placement, and cube placement. Each variant reuses its base-task demonstration while changing controlled object geometry, container or box-like instance, or spatial layout. The five-object basket task is a separate compositional track conditioned on four atomic-skill videos rather than a full-task demonstration.

C.4 SEED CONSTRUCTION AND PHYSICAL VERIFICATION

Deterministic candidate instantiation. Before environment initialization, each integer seed fixes the NumPy and Torch random states. Evaluation uses `demo_clean`: global background, lighting, camera, table-height, and clutter randomization are disabled, leaving only the task-coded object position, orientation, and model-ID fields listed below. Consequently, a fixed task implementation and seed reproduce the same scene. The task-dependent `candidate_seeds` field is an ordered pool, not a declaration that every listed candidate was instantiated. Configured pools use task-dependent integer ranges 1000–1099, 1000–1119, 1000–1199, 1000–1299, or 1000–5000, but these bounds describe the available search order rather than the number of executed probes. Candidates are probed in order, and collection stops as soon as 20 accepted seeds have been obtained for a task.

Orientation-based settling and acceptance gate. Each instantiated scene is reset, advanced for 2,000 physics steps, and then observed for 500 additional steps. During the final 200 observation steps, explicit object-orientation drift must remain within approximately 3° . This is an orientation-based settling check; the current selector does not impose a separate translational-drift threshold. Initial and post-probe object poses must also be finite. A seed receives `ready=true` only when its privileged probe, the planner result, terminal checker, gripper state, and task-specific physical outcomes all pass. A probe run that ends before producing a physical outcome is recorded separately and does not by itself classify the scene as physically infeasible.

Verification scope. All 280 retained manifest entries have a matching `ready=true` probe under a single verification mode: a complete `task.play_once()` execution with the full planner chain, `planner_success`, the terminal checker, gripper state, and the task-specific physical outcome all passing. Every retained seed therefore carries a verified expert execution from the robot home state through the terminal predicate, so each base-task instance is certified physically solvable end to end rather than merely reachable at the grasp endpoint. Across all tasks, 318 candidates were actually instantiated and probed before the per-task stopping rule was met, and 280 were retained. The resulting 88.1% is a conditional yield among candidates that were run, not an acceptance rate over the complete task-dependent candidate pools (Table 21).

Table 21: Conditional seed yield among candidates actually instantiated and probed. Selection visits each ordered pool only until 20 accepted seeds are collected, so unvisited pool entries are not part of the denominator.

Task or group	Probed	Retained	Yield (%)
Ten tasks whose first 20 probes passed	200	200	100.0
<code>place_cube_on_cube</code>	22	20	90.9
<code>place_single_cube</code>	22	20	90.9
<code>beat_block_hammer_right</code>	24	20	83.3
<code>lift_pot</code>	50	20	40.0
All 14 tasks	318	280	88.1

Task randomization and terminal predicates. Tables 22–25 report the task-coded scene fields and terminal checker conditions used by the base suite. Position ranges are in world-frame meters; tolerances are stated explicitly in millimeters. “Lifted” is measured relative to the corresponding initial object height. TCP denotes the tool-center point.

Table 22: Seed randomization and terminal predicates for direct-grasp tasks.

Task	Task-coded scene fields	Terminal success predicate
grasp_single_bottle	$x \in [0.20, 0.28]$, $y \in [-0.16, -0.04]$; yaw $\in [-22.5^\circ, 22.5^\circ]$	Target lifted > 50 mm; right gripper closed
grasp_single_bottle_upright	$x \in [0.20, 0.28]$, $y \in [-0.16, -0.04]$; fixed upright orientation	Target lifted > 50 mm; right gripper closed
grasp_single_cube	$x \in [0.18, 0.26]$, $y \in [-0.16, -0.06]$; yaw $\in [-45^\circ, 45^\circ]$	Target lifted > 50 mm; right gripper closed
grasp_single_pen	$x \in [0.18, 0.26]$, $y \in [-0.20, -0.10]$; yaw $\in [-30^\circ, 30^\circ]$	Target lifted > 50 mm; right gripper closed
grasp_pen_leaning_cube	Group $x \in [0.16, 0.22]$, $y \in [-0.24, -0.20]$; yaw $\in [-15^\circ, 15^\circ]$; fixed 30° lean	Pen lifted > 50 mm; right gripper closed

Table 23: Seed randomization and terminal predicates for placement and stacking tasks.

Task	Task-coded scene fields	Terminal success predicate
place_cube_in_bowl	Bowl $x \in [0.10, 0.14]$, $y \in [0, 0.04]$; cube $x \in [0.22, 0.27]$, $y \in [-0.17, -0.10]$; cube yaw $\in [-45^\circ, 45^\circ]$	Cube previously lifted ≥ 50 mm; center distance < 55 mm; relative height 15–95 mm; bowl displacement < 5 mm; contact; gripper open
place_cube_on_cube	Red cube $x \in [0.20, 0.26]$, $y \in [-0.14, -0.06]$; base cube $x \in [0.10, 0.14]$, $y \in [0, 0.04]$; yaw $\in [-45^\circ, 45^\circ]$	Red cube previously lifted ≥ 50 mm; top-surface overlap ≥ 2 mm; height error ≤ 12 mm; base displacement ≤ 12 mm; contact; gripper open
place_single_cube	Cube as in grasp_single_cube; target $x \in [0.10, 0.14]$, $y \in [0, 0.04]$	Cube previously lifted ≥ 50 mm; x/y error each ≤ 35 mm; height error ≤ 35 mm; target displacement ≤ 2 mm; contact; gripper open
place_single_bottle_upright	Bottle as in the upright-grasp task; target $x \in [0.03, 0.09]$, $y \in [0.07, 0.13]$	Bottle previously lifted ≥ 50 mm; x/y error each ≤ 40 mm; height error ≤ 35 mm; target displacement ≤ 2 mm; contact; gripper open

Table 24: Seed randomization and terminal predicates for functional-contact and tool-use tasks.

Task	Task-coded scene fields	Terminal success predicate
click_bell_right	$x \in [0.12, 0.26]$, $y \in [-0.18, -0.04]$; two bell model IDs	Gripper closed; contact point within 25 mm of the bell functional point on each of x and y , and within 30 mm on z
beat_block_hammer_right	Hammer $x \in [0.02, 0.06]$, $y \in [-0.10, -0.04]$; block $x \in [0.14, 0.22]$, $y \in [0.06, 0.13]$; yaw $\in [-0.5, 0.5]$ rad	Hammer lifted with contact by > 40 mm; functional-point x/y error each < 20 mm; physical hammer–block contact; gripper closed

Table 25: Seed randomization and terminal predicates for dual-arm tasks.

Task	Task-coded scene fields	Terminal success predicate
place_shoe	Shoe $x \in [-0.25, 0.25]$, $y \in [-0.10, 0.05]$; ten model IDs; samples with horizontal radius < 0.15 m excluded	x error < 50 mm; y error < 20 mm; absolute error of each quaternion component < 0.07 ; both grippers open
handover_horizontal_block	$x \in [-0.20, -0.15] \cup [0.15, 0.20]$, $y \in [-0.05, 0]$	Required handover state chain completed after vertical presentation; giver open and receiver closed; $z > 0.92$ m; object on receiver side; long axis within 20° of vertical
lift_pot	$x, y \in [-0.05, 0.05]$; yaw $\in [-22.5^\circ, 22.5^\circ]$; two pot model IDs	Pot $z > 0.82$ m; each TCP within 30 mm of its corresponding handle; upright dot product > 0.8

The shared target-placement predicate is implemented in `envs/_single_right_arm_place_target_task.py`; the remaining terminal predicates reside in their corresponding task environments. Deterministic initialization and the `demo_clean` policy are implemented in `envs/_base_task.py` and `task_config/demo_clean.yml`, while ordered candidate screening and both probe modes are implemented in `codex-robotwin-agent/scripts/select_safe_eval_seeds.py`.

C.5 METRIC ROBOT ACTION SPACE

The environment action space is

$$\mathcal{A} = \mathcal{A}_{\text{cam}} \cup \mathcal{A}_{\text{move}} \cup \mathcal{A}_{\text{rot}} \cup \mathcal{A}_{\text{grip}} \cup \mathcal{A}_{\text{dual}}. \quad (7)$$

Table 26 gives the exact robot-command parameterization. The interface exposes numerical Cartesian target changes rather than the legacy small/medium/large motion labels or low-level motor commands.

Table 26: Robot actions exposed to the MLLM. In dual-arm mode, every single-gripper numerical command must name the controlled arm.

Action family	Parameters	Semantics
World translation	$u \in \{x, y, z\}$, sign $\sigma \in \{-, +\}$, $d \in [1, 100]$ mm	Move the selected end effector by σd along a fixed world axis; the target is bounded by the robot workspace.
Local rotation	$r \in \{r_x, r_y, r_z\}$, sign σ , $\alpha \in [1, 90]^\circ$	Rotate about the selected current gripper-local axis using the right-hand rule; compensated waypoints keep GC approximately fixed.
Gripper aperture	open, close	Command the selected physical gripper without prescribing an object or contact state.
Synchronized dual translation	Independent (u, σ, d) for left and right <code>dual_gripper.open</code>	Plan and execute both world-axis targets in one RoboTwin move call.
Synchronized dual aperture	<code>dual_gripper.close</code>	Open or close both grippers in the same environment step.

These action families map to `gripper.move_world`, `gripper.rotate_local`, and `gripper.aperture` commands. Dual-arm episodes require `arm=left|right` for single-gripper commands and add the synchronized `dual_gripper` actions listed above. A fixed model-agnostic IK and trajectory layer attempts each metric target and returns binary planning feedback; it does not select the object, grasp region, arm role, or task waypoint. Section 3.5 defines this execution boundary.

C.6 BOUNDED ACTIVE CAMERA ACTION SPACE

The main evaluation exposes the camera actions in Table 27. Their semantic views are defined relative to GC, not to a privileged target-object location. Let c be the selected arm’s finger midpoint (or the midpoint of both arms), and let the sign of the lateral offset follow the selected arm side. Each semantic-view action moves the camera toward the listed nominal offset and orients it toward c ; the final pose is clamped to the camera workspace.

In single-arm tasks, the five semantic views target the active gripper. In dual-arm tasks, generic views target the two-arm midpoint, while `view_left_*`, `view_right_*`, and `view_both_*` explicitly select the observation target. Camera poses are clamped to a bounded workspace. The action changes only the observation viewpoint and never changes the world-frame meaning of robot actions.

We keep camera control discrete because continuous parameters would add control burden without making evidence acquisition more difficult. This interface deliberately evaluates task-conditioned viewpoint selection and bounded camera adjustment, not physical camera-arm motion. Semantic views and incremental camera actions directly update the simulated camera pose; camera reachability, collision, continuous trajectory cost, and calibration error are outside the current scope. Each

Table 27: Camera actions in the active-view protocol. Semantic viewpoints are gripper-centered; incremental translations and rotations are expressed in the current camera frame unless stated otherwise.

Action	Motion	Primary geometric evidence
view_topdown	$c + (0, 0, 0.68)$ m	World- XY displacement, planar yaw, target coverage
view_side	$c + (\pm 0.58, -0.02, 0.06)$ m	Height, table clearance, vertical overlap and insertion
view_front_side_45	$c + (\pm 0.44, -0.44, 0.18)$ m	Break front/side occlusion and horizontal projection ambiguity
view_side_top_45	$c + (\pm 0.48, -0.02, 0.48)$ m	Joint evidence for lateral alignment, height, and finger placement
view_oblique_45	$c + (\pm 0.42, -0.42, 0.42)$ m	Overall 3D configuration and selection of the next diagnostic view
move_left/right move_up/down	Translate 5 cm along camera left/right/up/down	Local de-occlusion and small framing changes
zoom_in/out	Translate 8 cm along camera forward/backward	Inspect or widen the local interaction region
yaw_left/right	Rotate 10° around world Z	Horizontal parallax and side exposure
pitch_up/down	Rotate 10° around camera-left axis	Vertical parallax and top/side exposure

camera action still consumes an environment step, and the agent instruction limits camera-only chains to at most five consecutive actions. Table 27 is the complete set of camera commands available in every reported active-view run; the passive control condition of Appendix C.7 exposes none of them.

C.7 PASSIVE FIVE-VIEW OBSERVATION PROTOCOL

The matched Qwen3.8-max comparison supplies five clean, predefined RGB images at each observation, in place of the single actively acquired image used by the canonical condition. The passive condition exposes no camera-action command. The textual observation block appears first, followed by image blocks in the same fixed order `topdown`, `side`, `front_side_45`, `side_top_45`, and `oblique_45`. The local session record stores these view labels, but the evaluated OpenAI-compatible API request includes no additional textual label beside an image; the fixed block order is therefore the model-visible cue to view identity. The observer uses a 55° vertical field of view, a 0.05-m near plane, and a 100-m far plane. The images contain no end-effector or target marker, segmentation mask, depth map, or debugging overlay. These camera parameters define the renderer but are not sent to the model as numerical calibration.

The passive five-view bundle is supplied after reset and robot actions at the same control points where the canonical protocol supplies its current image. The active model may then select further views through camera actions, whereas the passive model has no such command. Its API observation contains only the current five-image bundle; prior robot actions, planner outcomes, and textual feedback remain under the same bounded-history and compaction policy as the canonical evaluation. Robot-action semantics and environment transitions are also matched. The comparison therefore contrasts model-directed active evidence acquisition with passively supplied multi-view evidence.

C.8 DEMONSTRATION-CONDITIONED EVALUATION PROTOCOL

For each standard task, the main protocol supplies one RGB-only video of a successful expert execution. Before evaluation, the evaluated MLLM retrieves multiple sampled frames and writes a structured textual summary of visible constraints, including grasp region, finger placement, approach, posture, arm assignment, coordination order, and completion conditions. The video contains visible robot and object motion, and therefore conveys procedural information, but exposes no machine-readable expert action, EEPose, joint state, object coordinate, contact label, or semantic phase. The summary is task-level experience; it does not contain the metric solution for a live seed.

This task-level procedure video is distinct from the per-seed acceptance probes in Appendix C.4; those probe executions and their recorded trajectories remain evaluator-only and are never shown to the evaluated model. Demonstration interpretation and procedural abstraction are therefore part of the primary end-to-end construct, rather than fixed preprocessing.

For the primary model-specific-summary condition, this sequence is

$$D_\tau \xrightarrow[\text{multiple distinct RGB frames}]{\text{model inspection}} s_{m,\tau} \xrightarrow[\text{new seed}]{\text{re-grounding}} \{a_t, o_t\}_{t=0}^{t_{\text{end}}}. \quad (8)$$

At test time, the model receives the task instruction, generic safety and manipulation constraints, any task-specific semantic completion constraints, and its demonstration-derived summary. It must nevertheless reacquire the current scene, choose views, assign arms where applicable, and generate every numerical parameter. No oracle trajectory, target waypoint, target-object state, or task phase is supplied. The model-specific-summary condition is the primary end-to-end protocol. We additionally report a shared-summary condition as a controlled diagnostic that fixes the textual procedural context across models and therefore partially isolates downstream re-grounding and execution. Held-out variants reuse the base-task summary without a new video, while the composite basket task uses four independent atomic-skill videos and no full-task demonstration.

C.9 TRAJECTORY-BASED CAPABILITY DIAGNOSTICS

Terminal task success is deliberately strict, but it does not reveal whether a model failed to find the target, misunderstood the robot–object geometry, selected an unsuitable motion, or was unable to recover. We therefore augment the environment outcome with a trajectory-level diagnostic rubric. The rubric decomposes task competence into three first-level modules—*spatial perception and understanding*, *robot manipulation*, and *error recovery*—with three separately scored trajectory-level behavioral criteria in each module.

Evidence and annotation unit. The annotation unit is one complete episode. Each complete trajectory is independently reviewed by three annotators using the same episode record and rubric. Annotation proceeds in two stages: the three annotators first judge criterion applicability, and the majority decision fixes the episode-level gate; for every activated criterion, all three then record a binary behavioral judgment whose majority vote is the reported episode label. Annotators inspect the task instruction, the sequence of model-facing RGB observations, camera and robot actions, resulting scene changes, action validity, planner outcomes, and the terminal task state. The three independent judgments are retained in the annotation record, and the aggregation equations below operate on the episode-level labels used for reporting. Detailed positive and negative evidence rules and applicability conditions are provided in this appendix. Figure 3 gives a visual reference for the decision standard, with positive and negative trajectory examples for eight criteria and a scoring guide for FG. A model’s textual rationale can help identify its stated intent, but does not by itself establish a capability: the executed trajectory must contain corresponding behavioral evidence. Conversely, a single planner rejection is not automatically a planning failure, because a geometrically reasonable target may still be infeasible for the fixed execution layer. The judgment instead considers whether the model subsequently changes its estimate or repeatedly issues actions that remain inconsistent with the visible task.

Figure file missing in source package

figures/9ability_precontact_cropped.pdf

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Figure 3: Annotation reference for the nine trajectory-level capability criteria. In the eight trajectory panels, the upper row is a representative positive trajectory (green check) and the lower row is a representative negative trajectory (red cross). The FG panel gives its pre-contact analysis rule; final placement alone does not establish this behavior. The examples illustrate observable evidence, not hidden model states. For error detection (ED), online correction (OC), and post-failure adjustment (PF), the examples contain an observable error; error-free episodes remain NA.

For annotator $a \in \{1, 2, 3\}$, let $g_e^{(a)} \in \{0, 1\}$ indicate whether episode e contains an observable execution error. The shared recovery gate is the majority vote

$$g_e = \mathbb{I}\left[\sum_{a=1}^3 g_e^{(a)} \geq 2\right]. \quad (9)$$

If $g_e = 0$, ED, OC, and PF are all recorded as $\perp$ (reported as NA) and the episode is excluded from all three denominators. If $g_e = 1$, all three criteria are activated. For each activated criterion k , annotator a records $r_{e,k}^{(a)} \in \{0, 1\}$, where one denotes positive behavioral evidence and zero denotes that the behavior was not demonstrated. The episode-level label used for reporting is

$$r_{e,k} = \mathbb{I}\left[\sum_{a=1}^3 r_{e,k}^{(a)} \geq 2\right]. \quad (10)$$

The same majority-vote rule is used for the binary labels of the other criteria. AE carries an analogous assessable-ambiguity gate, but in the annotated runs every episode was judged to contain a task-relevant ambiguity under single-view observation, so the gate never fired and AE is scored over all annotated episodes; TL, SR, MS, MP, and FG are applicable by construction. Thus, an episode is not penalized for having no error, but an applicable capability is scored zero when at least two annotators find no corresponding behavioral evidence.

C.9.1 ANNOTATION RELIABILITY

Because the nine criteria are human-annotated, we report their agreement directly. The annotation record retains, for every activated criterion of every annotated episode, how many of the three annotators recorded positive evidence. Across the 3,360 annotated base episodes of the 12 primary conditions, this gives 29,232 applicable episode–criterion items and 87,696 individual binary judgments. Table 28 summarizes them. Terminal task success is determined solely by the environment

Table 28: Inter-annotator agreement over the nine trajectory criteria. n is the number of applicable episode-criterion items (three judgments each); ED, OC, and PF are scored only on episodes containing an observable error, hence their smaller n . Unanimity is the fraction of items on which all three annotators agree; every remaining judgment is a 2–1 split. Positive rate is the fraction of majority-vote labels equal to one, and drives the difference between κ and AC1 for near-ceiling criteria. Statistics cover the annotated base run of the 12 primary conditions; the held-out-transfer and long-horizon annotations are excluded here and contained no split votes.

Module	Criterion	n	Unanimity	Pos. rate	Fleiss κ	Gwet AC1
SPU	TL	3,360	97.41%	0.974	0.723	0.982
SPU	AE	3,360	92.77%	0.376	0.898	0.908
SPU	SR	3,360	94.70%	0.789	0.892	0.948
RM	MS	3,360	99.88%	0.988	0.968	0.999
RM	MP	3,360	97.08%	0.565	0.960	0.962
RM	FG	3,360	87.38%	0.321	0.797	0.856
ER	ED	3,024	96.40%	0.447	0.951	0.953
ER	OC	3,024	99.40%	0.124	0.982	0.995
ER	PF	3,024	96.76%	0.369	0.953	0.960
All criteria		29,232	95.69%	0.559	0.942	0.943
All except FG		25,872	96.77%	0.589	0.956	0.958

checker and is never annotated, so annotation reliability bounds the interpretation of the nine diagnostics and of subtask progress, not of any success rate reported in this paper.

All 1,259 disagreements are 2–1 splits, and every reported label matches the majority vote. Overall unanimity is 95.69%, with Fleiss $\kappa = 0.942$ (Fleiss, 1971).

For criteria whose positive rate is far from balanced, κ understates agreement, because a skewed marginal inflates the chance-agreement term. TL is the clearest case: annotators agree unanimously on 97.41% of episodes, but the near-ceiling positive rate yields $\kappa = 0.723$. We also report Gwet AC1, which is less sensitive to prevalence, and interpret the two together. FG has the lowest unanimity (87.38%, $\kappa = 0.797$) and accounts for 424 of the split votes (Gwet, 2008). Its individual scores warrant particular care when comparing voting rules.

Robustness to the aggregation rule. Majority voting is one of several defensible ways to collapse three judgments. We recomputed all nine criteria for all 12 primary conditions under two alternatives: a strict rule that credits a criterion only when all three annotators are positive, and a permissive rule that credits it when any annotator is positive. Averaged over the nine criteria, the largest within-condition range across the three rules is 6.25 points. Relative to majority voting, the strict rule decreases this mean by at most 3.93 points and the permissive rule increases it by at most 2.33 points. The induced orderings are similar (Spearman $\rho \geq 0.979$ and Kendall $\tau \geq 0.909$ for every pairwise comparison). The three strongest conditions under majority voting remain the three strongest under both alternatives. Individual criteria are more sensitive: GPT-5.6-sol leads FG under unanimity (62.5%), whereas Opus-5 leads under majority voting (77.9%). Thus stability of the nine-criterion means does not establish stability of every individual score or ordering.

Annotations are not restatements of the outcome. A concern for any trajectory-level rubric is that annotators may simply infer the labels from whether the episode succeeded. The record contradicts this. On the 1,680 annotated episodes for which the per-episode file also records the checker verdict, FG agrees with terminal success on 82.08% of episodes. There are 116 episodes with FG= 0 and checker success, and 185 with FG= 1 and checker failure. FG measures pre-contact analysis, so a positive label does not require completion, and success alone does not establish such analysis. These counts do not identify the cause of individual mismatches. TL and MS behave in the opposite way: they agree with terminal success on 31.1% and 30.7% of the same episodes, because more than 1,100 episodes are labeled positive on target localization and manipulation semantics while still failing, and *no* episode is labeled negative on either criterion while succeeding. Annotators that copied the outcome could not produce this pattern. The same asymmetry is itself a finding: the visual and semantic prerequisites are usually satisfied, and the failures accumulate downstream.

What the record does not support. The retained counts are per-episode tallies rather than per-annotator labels, so statistics that require annotator identity—pairwise Cohen κ , per-rater bias estimates, and Cochran’s Q —are not recoverable and we do not report them. The concordance figures above are computed on the subset of annotation files that carry a machine-readable success column (1,680 of the 3,360 annotated episodes); the agreement statistics in Table 28 use all 3,360.

C.9.2 SPATIAL PERCEPTION AND UNDERSTANDING

Target perception and localization. This criterion records whether the executed behavior identifies the task-relevant object and localizes it well enough to guide subsequent behavior. Positive evidence includes end-effector motion directed toward the correct object, camera actions that keep the target in the region of interest, and later operations organized around that target. A negative judgment requires clear contrary evidence, such as sustained motion toward an unrelated object or workspace region, grasping a distractor, or repeatedly operating far from the target. An isolated exploratory motion is not sufficient to infer a localization failure.

Informative active exploration. This criterion records whether the executed behavior is consistent with actively acquiring and using visual evidence that resolves a task-relevant ambiguity. An exploration opportunity is annotated when the current observation is judged insufficient for a spatial or geometric fact needed by the next action, such as object orientation, depth, the contact region, the relative pose of two arms, or the visibility of a container opening. Uncertainty about whether an executed action achieved its intended effect is assessed under error detection rather than under this criterion.

An episode receives positive evidence only when all three pieces of behavioral evidence are present: (i) the model selects a semantic view or camera motion in response to the ambiguity; (ii) the resulting observation exposes information that reduces that ambiguity; and (iii) a subsequent action or state estimate is consistent with the new observation. A redundant or task-irrelevant viewpoint, or a view change whose observation does not reduce the ambiguity, is scored 0. Visiting an informative view and then immediately switching to an ineffective view without using the acquired evidence is also scored 0; a later irrelevant view does not erase positive evidence when the informative observation was already used. Episodes with no assessable task-relevant ambiguity would be marked $\perp$, but no annotated episode fell in this category: under single-view observation some task-relevant geometric fact was always judged underdetermined. The criterion measures trajectory evidence of evidence acquisition and use, not terminal task success.

Robot–object spatial relations (SR). This criterion records whether the trajectory is consistent with correct interpretation of relative height, distance, direction, and alignment between the gripper and the target. Positive evidence occurs when successive translations or rotations reduce the visible robot–object offset and it initiates contact, grasping, or release at a geometrically plausible state. Clear failures include closing the gripper while it remains visibly above or beside the object, descending at an incorrect lateral position, confusing image-plane direction with world-axis motion, or declaring alignment while a substantial separation remains.

C.9.3 ROBOT MANIPULATION

Manipulation-semantics understanding. This criterion records whether the agent’s behavior is consistent with correct use of the available control commands and selection of task-relevant actions. Positive evidence is a purposeful use of world-axis translation, local-axis rotation, aperture, or dual-arm commands to produce the intended physical effect. Evidently meaningless rotations, uncontrolled large motions, repeated open/close commands without a contact opportunity, or actions fundamentally unrelated to the task are inconsistent with the intended manipulation semantics. Numerical imprecision alone is assessed under spatial understanding rather than automatically counted here; FG separately records pre-contact analysis.

Goal-directed manipulation planning. This criterion evaluates whether the agent constructs a coherent, goal-directed metric action sequence from the current state to the desired interaction state. Evidence includes an appropriate ordering of approach, alignment, grasp/contact, lift or actuation, transport, placement, release, and retreat, using only the phases required by the task. Intermedi-

ate actions should progressively reduce relevant spatial error or establish a necessary precondition. Large purposeless excursions, repeated movement in unrelated regions, an approach direction incompatible with the intended contact, or a sequence whose phases cannot lead to the goal constitute negative evidence.

Fine-grained pre-contact analysis (FG). This criterion records whether the model performs fine-grained spatial analysis as it approaches an intended contact. Relevant details include fingertip placement, gripper orientation and aperture, contact location, approach or placement height, and relative arm geometry. Positive evidence is a task-specific assessment of these details reflected in pre-contact decisions and actions. A generic instruction to manipulate carefully without corresponding behavior is insufficient. Every episode is scored; absence of such analysis, including failure to reach an opportunity for it, receives 0. The criterion does not require successful operation completion. A later miss, unsuitable release, or terminal failure does not by itself negate demonstrated pre-contact analysis, and terminal success alone does not establish it.

C.9.4 ERROR RECOVERY

Error detection (ED). After an observable error occurs, ED is positive when subsequent behavior shows that the agent no longer treats the preceding action as successful—for example, it stops the current sequence, obtains another view, rechecks the grasp, or begins a materially different adjustment. Continuing the same plan while ignoring a visible miss, dropped object, unstable placement, or repeated planner rejection is negative evidence. Error-free episodes are NA for ED.

Online correction (OC). After an observable error occurs, OC is positive only when the agent notices and corrects a developing spatial deviation before it becomes an explicit operation failure. Examples include adjusting lateral alignment or gripper orientation before closing, re-centering an object during transport, and restoring bimanual levelness before the payload slips. If the error is allowed to progress into failure, or no timely corrective action is shown, OC is 0. Error-free episodes are NA for OC.

Post-failure adjustment (PF). After an observable error becomes an explicit failure, PF is positive when the agent re-evaluates the new state and changes its observation strategy, geometric hypothesis, approach, or numerical parameters before retrying. The recovery attempt need not ultimately complete the task: a clearly targeted, materially revised attempt is positive behavioral evidence. Repeating the failed action without a meaningful change, persisting with an invalid stale plan, or terminating despite a recoverable state is negative evidence. If an error occurs without a post-failure adjustment, PF is 0; error-free episodes are NA. OC and PF are stage-specific rather than mutually exclusive: one episode may receive positive credit for correcting a developing deviation and later making a separate post-failure adjustment.

Aggregation. Let

$$E_m = \{e : g_e = 1\}.$$

For each error-recovery criterion $k \in \{\text{ED}, \text{OC}, \text{PF}\}$, the reported score is the positive rate over this shared error set,

$$C_{m,k} = \frac{\sum_{e \in E_m} \mathbb{I}[r_{e,k} = 1]}{|E_m|}, \quad (11)$$

where $|E_m|$ is the error-episode denominator used for the score. For the other six criteria, the denominator remains the number of episodes with a valid opportunity to assess that criterion. Each first-level module is summarized by the equal-weight macro-average of its three constituent criterion scores; these module values are descriptive summaries, not an overall capability score. We retain all nine criterion scores in addition to the three module summaries so that a high value in one capability cannot conceal a distinct failure mode in another. We do not collapse all nine criteria into a single scalar because their applicability sets and statistical reliability differ. If a capability has no applicable episode, its score is reported as NA rather than zero and it is not silently included in a module average.

Subtask progress for complex tasks. For a multi-stage task, terminal success is supplemented with a task-specific checklist of observable subgoals, such as acquiring the correct object, completing a required orientation change or handover, reaching the destination, and establishing a stable final state. A subtask receives credit only when its required state transition is visible in the trajectory; merely issuing the intended command or obtaining a successful planner return is insufficient. Subtask completion is reported separately from terminal success and from the nine general criterion scores. Let $h_{m,\tau}$ and $n_{m,\tau}$ denote the completed and scored checkpoint instances, respectively, for model m on task τ . Across the base benchmark, we use the checkpoint-level micro-average

$$P_m^{\text{sub}} = \frac{\sum_{\tau=1}^{14} h_{m,\tau}}{\sum_{\tau=1}^{14} n_{m,\tau}}. \quad (12)$$

Thus, each checkpoint instance is equally weighted, while tasks with more checklist stages contribute proportionally more instances. This preserves partial progress in long-horizon and bimanual tasks without relaxing the benchmark’s environment-defined success criterion.

D AGENT HARNESS IMPLEMENTATION DETAILS

We evaluate every model through the same model-agnostic agent harness rather than attaching a learned robot policy or action head. The harness is an independent Python implementation of a persistent tool-using agent loop, specialized for interactive robot evaluation. It separates model inference, conversation state, action validation, environment execution, and run logging so that the MLLM is the only model-dependent component of the control stack.

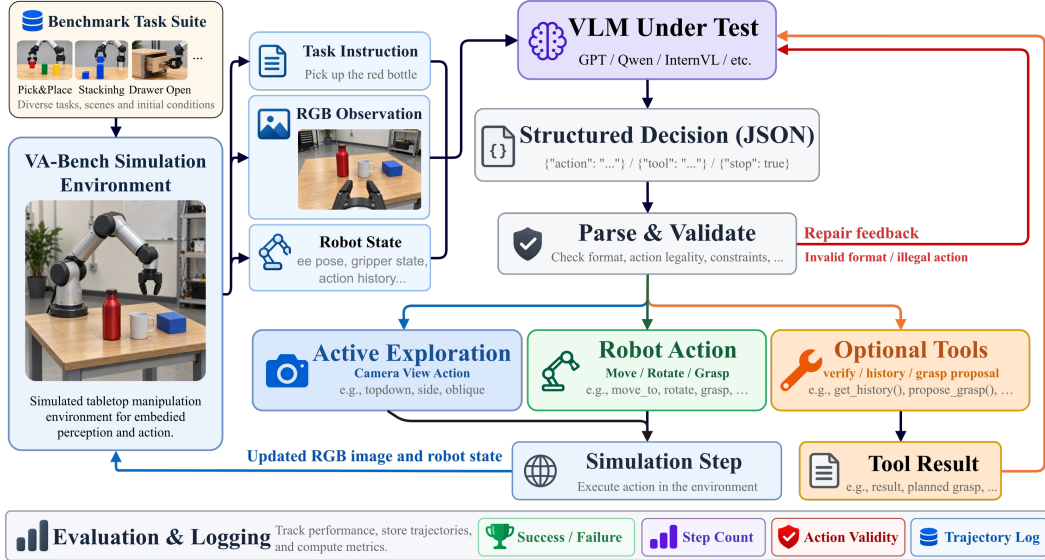


Figure 4: Schematic of the shared agent harness. Structured model decisions are validated and routed to camera control, robot execution, or non-environment tools, with observations and feedback returned to the model. Task, model, and command examples are illustrative. In the primary evaluation, robot actions are metric translations, rotations, and gripper open/close commands; the illustrated grasp-proposal option is not used. Camera actions change only the observation pose, and non-environment tools do not advance the simulator. Demonstration-derived summaries also condition the model but are omitted from the diagram.

D.1 PERSISTENT DECISION LOOP

At each outer control decision, the harness sends the current observation payload together with the retained conversation state and asks the MLLM for one JSON object. The object denotes a non-environment tool call, a robot action, a camera action when enabled by the observation protocol, or

a stop decision. Tool results are returned to the same thread and may trigger another model call; they do not advance the simulator. Robot and camera actions pass harness-level schema and numerical-range validation before final environment-side arm, validity, and workspace checks. Malformed JSON and invalid decisions are returned to the model for a bounded number of repair attempts, and tool-call iterations are likewise bounded. The harness provides adapters for OpenAI-compatible Chat Completions and Responses endpoints and Anthropic Messages endpoints, normalizing their outputs into this common JSON protocol rather than relying on a provider-specific robot tool interface. The canonical active-view condition exposes camera actions. The passive five-view condition removes camera actions from the model-visible JSON vocabulary because its five predefined views are supplied directly; the robot execution stack is unchanged.

The same conversation thread is retained across the entire episode. Both conditions use the standard context-length limit and compaction policy; prior robot actions, planner outcomes, action-validity feedback, and textual messages remain in the retained history. Each active API request contains only the current RGB image, whereas each passive request contains only the current fixed-order five-image bundle. Earlier visual payloads are not re-sent as the current observation. Compaction leaves the append-only session record intact while bounding the model-visible context.

For the primary base-task terminal-success comparison, the harness repeats the fixed 20-seed suite in three independent runs for each of the 12 evaluated models. Run identity is retained in the session record, and aggregation is performed only after computing the task-macro-average within each run. The passive five-view control, held-out transfer, composition, and trajectory annotation analyses retain their separately identified matched single-run protocols.

D.2 MODEL OBSERVATIONS AND ACTIONS

After reset and after every robot action, the harness acquires the next RGB payload together with robot proprioception and retained robot/action history. The active API sends one current image; the passive API sends only the current control point’s fixed-order five-image bundle. An active camera action may return a newly selected view without advancing physical state; the passive condition has no such command. Context limit, compaction policy, and environment transitions are otherwise shared. Proprioception includes gripper state, end-effector pose, the midpoint between the physical fingers (GC), and the current gripper-local axes expressed in the world frame. The model also receives the preceding action’s validity and a binary trajectory-planning outcome. It does not receive the target object’s coordinates or pose, segmentation masks, oracle grasp points, contact state, or task waypoints. The fixed motion-planning layer necessarily accesses simulator state required for inverse kinematics and trajectory generation, but this privileged state is not exposed to the MLLM.

The robot interface contains world-frame Cartesian translation increments, gripper-local rotation increments, and open/close commands for one or two arms. Numerical translations and rotations are bounded; Cartesian targets are clamped to the configured robot workspace before execution. For a local rotation, the adapter transforms the requested gripper-local axis into the world frame, compensates the end-effector position so that GC remains approximately fixed, and decomposes large rotations into planner waypoints of at most 10° . A fixed inverse-kinematics and trajectory layer then attempts the requested motion. It may reject an unreachable trajectory, but does not select the task object, grasp, or semantic waypoint for the model.

When enabled, the camera interface combines five gripper-centered semantic viewpoints with bounded translation, zoom, yaw, and pitch actions. These actions directly update the simulated camera pose: they do not model a camera arm’s reachability, collision constraints, or continuous path cost. Camera and robot actions both consume the episode step budget, whereas history, self-geometry, and demonstration-retrieval tools consume inference and tool budget without advancing the physical episode. We therefore characterize this interface as bounded task-conditioned viewpoint selection, not unconstrained physical camera trajectory planning or camera-arm control.

D.3 DEMONSTRATION AND PROMPT CONDITIONING

In the demonstration-conditioned protocol, the evaluated model first inspects sampled frames from an RGB-only video of a successful execution and writes a structured task summary. This track exposes no machine-readable expert action, end-effector pose, joint state, object coordinate, con-

tact label, or semantic phase. The visible robot and object motion nevertheless conveys procedural supervision, including operation order and approach strategy; the protocol removes direct numerical action supervision rather than making the demonstration action-free. A learning gate requires the model to inspect multiple distinct frames before its summary can be stored. Model-specific-summary and shared-summary tracks allow the effect of self-distilled experience to be studied separately. The model-specific-summary condition is the primary end-to-end protocol, whereas the shared-summary condition is a controlled diagnostic rather than an interchangeable input for cross-model ranking.

The model is additionally conditioned on the task instruction, generic manipulation and safety constraints, and task-specific profiles where needed for roles or completion conditions. Thus, planning remains open-ended in the chosen observation–action sequence and numerical parameters, but it is not an unscaffolded discovery process. No low-level ground-truth trajectory, oracle waypoint sequence, or target-object state is included in this scaffolding.

D.4 EXECUTION RECORDS AND OUTCOME

The adapter stores the clean RGB frames delivered to the model separately from annotated debugging and recording views. Messages and events are written to append-only JSONL files, while image artifacts are stored separately and referenced by the session record. Per-step records include the selected action, the model-provided reason, robot self-geometry, action validity, and the planner outcome. An action is dispatched through the adapter to RoboTwin, which computes the next observation and invokes the task’s own success check. Consequently, reported success is the environment-defined, physics-simulated task outcome rather than a visual judgment made by either the MLLM or the harness. As emphasized in Section 1, this is an end-to-end system measure whose outcome also depends on the fixed planner, action interface, and simulator dynamics.